ZUGZWANG FOUNDATION · EXPERIMENT PHASE

Product Specification

A Reputation Market protocol — a platform for debate over the questions facing humanity, where participants stake reputation rather than capital, built to test whether true knowledge prevails over manipulative capital across the course of a debate, and so shape what happens next in humanity's favour.

DOCUMENT	SPEC.1 · v1.0.0-draft
STATUS	Draft (awaiting review)
AUTHORS	The Zugzwang Authors
LAST UPDATED	2026-05-03
READER	Claude Code (primary), human reviewers (secondary)
SOURCE	github.com/zugzwang-foundation/experiment

READER CONTRACT

> ***Reader contract:*** *SPEC.1 is the source of truth for what the Zugzwang Experiment does and why. Architecture lives in SPEC.2. Build contract lives in* `CLAUDE.md` *and* `AGENTS.md`*. Architectural Decision Records in* `docs/adr/` *override. When this document and code disagree, this document wins until updated by an ADR.*

CONTENTS

§0 Document Metadata

Thesis relevance: (b) operationally enabling.

- **Version:** 1.0.0-draft (semver; bump major on invariant changes)
- **Last updated:** 2026-05-03
- **Authors:** The Zugzwang Authors
- **Status:** Draft (promotes to Approved on v1.0.0 lock)
- **Related contracts:** `CLAUDE.md`, `AGENTS.md`, `docs/specs/SPEC.2.md` (architecture, forthcoming), `docs/adr/`
- **Reader:** Claude Code (primary), human reviewers (secondary)
- **Source decisions absorbed:** `spec1_gap_decisions.md` (75 rows resolved 2026-04-30); `cluster_a_ratify.md` (19 defaults); `cluster_b_decisions.md` (privacy/identity, 5 rows); `cluster_c_decisions.md` (visibility/brand/conclusion, 6 rows); `cluster_d_decisions.md` (operational edges, 4 rows); `cluster_e_decisions.md` (E1+E2 moderation, provisional); `cluster_launch_surfaces.md` (I1/I4/I5/I6); `Cluster_B` (A2 award formula, B1 lifecycle, B5 admin role); `Meta-rule__Prior-art_reference_policy`.
- **Parked outside SPEC.1 scope:** A6 → `FOUND.1` (pre-registered hypothesis); J1 → Hrishikesh sole owner (refusal: spec does not invent market content); J9 → number-tuning pass.

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§1 One-Paragraph Product Description

Thesis relevance: (a) directly testing $K \times n > C$.

The Zugzwang Experiment is a Reputation Market — a platform for debate over the questions facing humanity, where participants stake reputation rather than capital, built to test whether true knowledge prevails over manipulative capital across the course of a debate, and so shape what happens next in humanity's favour.

The user is anyone, anywhere, who holds a position on a contested question facing humanity and is willing to put their reputation behind it. The problem the protocol tests is that in current opinion-formation systems, capital systematically outweighs knowledge: a well-funded actor with weak claims can swamp informed actors who lack the money to push back, and by the time the world resolves the question, the prevailing narrative has already shaped what gets decided.

Every debate on Zugzwang takes the same shape. A *Ruling Party* commits to one side of a contested question and stakes reputation on it; an *Opposition* commits to the other and stakes equally; an *Audience* reads, watches, and forms a view, with the freedom to cross into either side as the debate persuades them. The two staked sides are structurally symmetrical — neither is presumed correct, neither carries more institutional weight.

The mechanism is a prediction market, but the unit of stake is reputation. Prediction markets are well-studied for the way game theory pulls private information into a public signal: a participant who believes the market price is wrong faces an incentive to act, because acting on a true belief is rewarded with reputation, and acting on a false belief costs them reputation. The aggregate of these individual decisions is a price that reflects what the staking population collectively believes, weighted by how much each participant is willing to risk. Zugzwang inherits this property and binds it to reputation. A staked position must be accompanied by an argument; the record of who said what under what stake is permanent and public; and the visibility of stake-backed correctness draws further informed participants in, growing the population of knowledgeable voices over the course of the debate.

Zugzwang's wager is that this combination — reputation as the staked unit, argument as a precondition for staking, an immutable record as the substrate — produces a price signal in which the informed-and-staking population dominates noise, capital, and inertia. The protocol claims that effective knowledge eventually overtakes the cost of manipulative capital, and that this overtaking happens early enough in the debate window to matter — before the world acts on whatever signal was available at the moment of decision. Success is when this race — knowledge against capital, knowledge against time — is observably won across debates whose outcomes shape what happens next.

§2 Glossary

Thesis relevance: (b) operationally enabling.

TERM	DEFINITION	CODE IDENTIFIER
Market	A binary YES/NO question with a deadline and a resolution criterion.	`markets` table, `Market` type
Bet	An atomic stake on a side of a market, accompanied by a comment.	`bets` table, `Bet` type
Comment	The textual / image argument attached to a bet, or a reply to one.	`comments` table, `Comment` type
Reply	A comment whose `parent_comment_id` is non-null. Inherits the *replier's* current side at reply-time.	`comments.parent_comment_id`
Side	YES or NO. The market's two share types.	`comments.side_at_post_time`, `bets.side`
Position	A user's net share holding in a market. Computed from the bet ledger.	derived; not a column
Pool	The CPMM share reserves for a market. Counterparty to every user trade.	`pools` table
Pool seed	A `pool_seed` Dharma flow from the admin account into the market pool at market creation.	`dharma_ledger.tag = 'pool_seed'`
Pool unwind	A `pool_unwind` Dharma flow from the pool back to the admin account at resolution or void.	`dharma_ledger.tag = 'pool_unwind'`
Dharma	Non-transferable reputation score. Single fluid unit; same instrument is staked, won, and lost.	`NUMERIC(38,18)` column on user rows; flows in `dharma_ledger`
Daily Allowance	A per-user, use-or-lose Dharma credit accruing once per UTC day.	`daily_allowance_events` table

TERM	DEFINITION	CODE IDENTIFIER
Pseudonym	The user's auto-assigned public display name of the form `<Colour><Animal><Number>` (e.g., `RedFox072`). Permanent. Not user-chosen, not user-editable. Not an alias for a real-world identity.	`users.pseudonym`
PFP	Profile picture: a pre-generated illustration of the user's `(colour, animal, number)` tuple, served from CDN. Coherent with the pseudonym. Permanent.	`users.pfp_filename`
Identity Pool	The pre-generated bank of `(colour, animal, number)` tuples and matching PFP images consumed by signup. Filled pre-launch via the asset pipeline (§13 F-AUTH-3); FIFO-consumed at signup; never replenished except by re-running the pipeline operationally.	`identity_pool` table
Track A	Moderation track: auto-ban category (CSAM, sexual minors, NSFW, adult imagery). No admin in loop.	`mod_actions.action ∈ {csam_blocked, nsfw_auto_banned}`
Track B	Moderation track: admin-review category (graphic violence, threats, hate, harassment). Admin in loop.	`mod_actions.action ∈ {flagged, approved, blocked}`
Track C	Moderation track: below threshold. Posts normally.	absence of `mod_actions` flag row
In / Flipped / Exited	The three-state marker on every comment, reflecting the *author's current position* relative to the comment's *frozen post-time side*.	derived from `bets` + `comments.side_at_post_time`
Open / Closed / Resolving / Resolved / Voided / Frozen	The six market lifecycle states. See §6.	`markets.state`
Banned	A user-account state. Existing positions ride to resolution; no new bets, comments, allowance accrual, or appeal.	`users.banned_at` (non-null = banned)
Mod-action log	Append-only log of every Track A and Track B action.	`mod_actions` table

TERM	DEFINITION	CODE IDENTIFIER
Admin-events log	Append-only log of every admin-initiated state-changing action.	`admin_events` table
K_eff	Effective knowledge present in a market's price at time t. K_eff(t) = $K_0 \cdot n(t) \cdot \sigma(t)$ per the whitepaper.	computed; surfaced via `/k-eff` dashboard
Zugzwang Condition Z	The thesis success criterion: $t^* < T$, where t^* is the time at which K_eff overtakes manipulative capital. Per whitepaper.	computed; reported in conclusion dataset
Admin / MM	The single operational identity (Hrishikesh) that creates markets, seeds pools, resolves markets, and operates moderation via the Admin Control Centre (§15). **Admin does not bet, comment, vote, or hold positions** — per `B5`, enforced structurally: admin authenticates via F-AUTH-ADMIN, has no `users` row, and therefore has no participant identity from which to act.	`admin_sessions` table; `ADMIN_EMAIL` env var; `admin_events.actor_id = 'admin-singleton'`
Admin Control Centre	The internal-facing UI consolidation of admin operations (§15). Hub at `/admin` plus inline admin-only affordances on public pages. Admin-only at every surface; non-admin requests are rejected at the auth middleware.	`/admin/*`, inline affordances
Brand-account	Zugzwang-owned social-platform accounts that propagate market activity outward. AI-agent-curated, admin-reviewed.	external; see `K2` and Social workstream
Review queue	Admin queue surface for Track B flagged comments and AI-generated brand-account posts.	`/admin/moderation`, `/admin/social-queue`
CPMM	Constant-product market maker. The pricing rule for share trades. Standard form: `x · y = k`.	`src/server/markets/cpmm.ts`
Stake	The Dharma amount committed to a bet. Bought back on sell; settled on resolution.	`bets.stake`

TERM	DEFINITION	CODE IDENTIFIER
Slippage	The price impact of a single trade against thin liquidity.	computed pre-confirm
Friendly-fire	Asymmetric upvote/downvote mechanic: a user upvotes only opposite-side comments and downvotes only same-side comments. Display-only — never a Dharma flow, never affects state, but does enter the §9 ranking function as one input among others.	`friendly_fire_events` table
Ranking function	Open-source, deterministic, universal function that orders top-level comments in the debate view. Lives in `RANKING.md`. Inputs include stake-at-post-time, friendly-fire counts, author Dharma, age, reply count. Audit-able from public dataset; not an audit event itself.	`RANKING.md`, `src/server/ranking/`
Single-side rule	Per-market constraint: a user holds a position on at most one side at a time. Enforced at the `(user_id, market_id)` position layer. To switch sides, exit fully then re-enter via a fresh F-BET-1 with new comment.	enforced in `src/server/bets/place.ts`

Drift between this glossary and code identifiers is a bug — a column rename, a type alias, or a route name change must update this section in the same PR.

§3 Goals and Non-Goals

Thesis relevance: (a) directly testing K × n > C.

3.1 Goals (numbered, testable)

- **G1.** Test the Zugzwang Condition Z: produce a public dataset across the experiment window from which K_eff(t) and its components can be measured directly, such that the informed-and-staking population — on whichever side of the question they sit — produces a price signal that prevails over noise, capital, and inertia. *(Maps to acceptance test* `dataset-zugzwang-condition-derivable`*.)*
- **G2.** Release the full archive — markets, bets, comments, Dharma ledger — as a public dataset on November 6, 2026. *(Per* `H1`*. Maps to* `archive-bundle-released-on-time`*.)*
- **G3.** Surface a live K_eff dashboard during the experiment, accessible authenticated and anonymous, refreshing throughout the window through the conclusion event. *(Per* `J5`*. Maps to* `k-eff-dashboard-live-throughout`*.)*
- **G4.** Enforce bet+comment atomicity at the database transaction layer such that no observable state contains a bet without its comment, or vice versa. *(Per* `INV-1`*. Maps to* `bet-comment-atomicity` *test family.)*
- **G5.** Make commentary mandatory at every market entry: the entry comment is the user's first stake-attached argument, side-frozen to the entry position. *(Per* `INV-1` + `INV-3`*. Maps to* `entry-bet-requires-comment`*.)*
- **G6.** Preserve pseudonym-stable reputation: a user's Dharma trajectory and comment record persist across the experiment window under a pseudonym chosen at signup. Pseudonym scrub on right-to-erasure replaces the display name with a permanent placeholder; the ledger and content remain. *(Per* `H2`, `D8`*. Maps to* `pseudonym-scrub-preserves-ledger`*.)*
- **G7.** Bound the admin role: the admin authenticates separately, creates markets, seeds pools, resolves, and moderates Track B. Admin cannot place bets, post comments, or hold positions — enforced structurally, not by runtime check: admin has no `users` row (per `B5`, F-AUTH-ADMIN). *(Maps to* `admin-not-participant` *test family.)*
- **G8.** Make the public dataset sufficient for *post hoc* analysis of propagation dynamics — including the rate at which informed participation grows (dn/dt) and how that growth relates to stake-backed correctness — by including timestamps on every bet, comment, and admin-events row already captured by §16.4. No live propagation dashboard ships; the goal is satisfied by the data being analysable downstream by anyone with the archive. *(No new flows, no new tables. Maps to* `dataset-propagation-derivable`*.)*

3.2 Non-Goals (high-level — full out-of-scope catalogue in §18)

- **NG1.** No testnet or mainnet scope. Web2 only. No blockchain primitives.
- **NG2.** No native mobile apps. Responsive web only.
- **NG3.** No end-to-end encryption. Hosted experiment.

- **NG4.** No federation. Single deployment, single domain.
- **NG5.** No per-user posting integrations to external platforms. Brand-account propagation is admin-curated only.

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§4 Personas and Primary Use Cases

Thesis relevance: (a) directly testing $K \times n > C$.

Every contested question has two sides and an audience. The protocol takes no position on which side carries truth, capital, or knowledge — none of those is a stable property of a side. They are properties of the debate as it unfolds, and they can flip. The personas below name positions in a debate, not identities of people. A participant is the Ruling Party in one market, the Opposition in another, part of the Audience in a third — and may switch role within a single market as their position changes. The admin is operational, not a persona, per `B5`.

The Ruling Party

The side whose claim, at this moment, is where the price sits. They may have arrived first, carry more stake, or simply occupy the position price discovery has converged on so far. None of this implies they are right, knowledgeable, or well-funded. Each stake they place enters at a price that already reflects their position, so the marginal information they add is small — until the price moves against them, at which point they are no longer the Ruling Party. The protocol serves them by making the dominant claim something that has to be defended in argument and stake rather than merely asserted, by recording what they said and when so a correct dominant-side call compounds reputation, and by ensuring challenges are paid for in stake rather than noise.

The Opposition

The side whose claim, at this moment, is against the price. They may have arrived later, carry less stake, or occupy the position price discovery has not yet converged on. None of this implies they are right, knowledgeable, or under-funded. Each stake they place enters at a price disfavoured to their side, so a correct call against the price pays disproportionately — until the price moves their way, at which point they become the Ruling Party and the asymmetry inverts. The protocol serves them by rewarding correctness against the price in reputation that compounds across markets under a stable pseudonym, by making their argument visible and stake-weighted so the contrary case is legible, and by preserving an audit-able record so a correct call does not get retconned by the prevailing narrative.

The Audience

Anyone reading without yet committing a position on a given market — unauthenticated, or authenticated but not staked here. Their attention is what the $K \times n > C$ race is *for*: the protocol's job is to surface price and arguments legibly enough for them to either join a side as an informed participant, or walk away with a credible read of where the question probably sits. Whether and how fast they convert is what the propagation-dynamics signal in the dataset measures. The protocol serves them by giving unobstructed read access without a login wall, by framing every market clearly as Ruling Party versus Opposition with labels that describe *current price position* and may flip as they read, by surfacing the K_eff trajectory and stake-weighted comments so either side's case is inspectable, and by releasing a credible, downloadable dataset at the end so the experiment's claim about reality is independently checkable.

§5 The Hard-Locked Invariants

Thesis relevance: (a) directly testing K × n > C — this is the spine.

This section mirrors `CLAUDE.md` §2.1–§2.4 verbatim in semantics. It exists in SPEC.1 because it is the gating contract for every code path Claude touches; drift between the two files is a bug fixed in the same PR. Conservation of Dharma (every trade is a flow between user and pool, no synthetic mint) and audit-trail immutability across `admin_events` and `mod_actions` are real and important rules — they are *enforcement* layers on these invariants, lived out in §10, §11, §15, and §16.4. They are not themselves §5 invariants. The closed set of invariants is four.

INV-1 — Bet ↔ comment atomicity

- **Statement.** A bet row and its associated comment row are either both persisted or neither is. There is no observable state in which a bet exists without its comment, or vice versa.
- **Rationale.** Mandatory commentary is the product's thesis. Allowing silent bets re-creates a generic prediction market.
- **Enforcement.** Single Postgres transaction wrapping both inserts. `bets.comment_id` is `NOT NULL` with foreign key. `POST /api/bets` (or the corresponding Server Action) without a `commentId` returns 400.
- **Test assertions.** `tests/server/bets/atomicity.test.ts`:
  - `it("rolls back the bet when the comment insert fails")`
  - `it("rolls back the comment when the bet insert fails")`
  - `it("rejects API calls missing either field with 400")`
  - `it("rejects bet inserts where comment_id is NULL with a constraint error")`
- **Failure mode.** Silent corruption: bets exist without commentary, thesis violated, dataset compromised. Recovery requires migration replay from last clean snapshot.
- **Code paths.** `src/server/bets/place.ts`, `src/app/api/bets/route.ts`, every Server Action that creates a bet, all admin tooling that synthesises bets (none in v1).

INV-2 — Dharma is non-transferable

- **Statement.** Dharma moves only as a market mechanic — staking on a bet, settling on resolution, or seeding/unwinding a pool. There is no user-to-user transfer.
- **Rationale.** Reputation that can be bought, gifted, or laundered is not reputation. The K × n term collapses if Dharma is fungible across identities.
- **Enforcement.** No `dharma_transfer` table by design. Every `dharma_ledger` row carries a `source` tag in a fixed enum: `bet_stake`, `bet_settle`, `daily_allowance`, `pool_seed`, `pool_unwind`, `correction_reverse`, `correction_apply`, `void_refund`. No "send Dharma" UI surface. No admin override that moves Dharma between accounts except via a resolution event.

- **Test assertions.** `tests/server/dharma/non-transferable.test.ts`:
  - `it("rejects any direct user-to-user dharma write")`
  - `it("requires a tag from the fixed enum on every ledger row")`
  - `it("admin pool_seed and pool_unwind flow account ↔ pool, never user → user")`
- **Failure mode.** Reputation laundering. Sybil farms purchase or transfer Dharma to game leaderboard and K_eff, invalidating the experiment.
- **Code paths.** `src/server/dharma/*`, `src/server/markets/pool.ts`, every code path that produces a `dharma_ledger` row.

INV-3 — Side is frozen at comment-time

- **Statement.** A comment inherits the author's market position at the moment the comment is posted. If the author later flips, exits, or re-enters, the comment's side label does not change. Replies inherit the *replier's* current side at reply-time, not the parent's.
- **Rationale.** Comments are stake-backed arguments; their meaning is bound to the side the author was on when they spoke. Allowing post-hoc reassignment converts the debate view into a self-rewriting record.
- **Enforcement.** `comments.side_at_post_time` is non-null and never updated after insert. A row-level rule rejects updates to that column. The author's *current* position is computed live on read and surfaces as the **In / Flipped / Exited** marker per `B1`. A user with zero current position cannot post a new comment (`A1`); their prior comments remain visible with the **Exited** marker.
- **Test assertions.** `tests/server/comments/side-frozen.test.ts`:
  - `it("preserves comment side after author flips position")`
  - `it("preserves comment side after author exits position")`
  - `it("inherits replier's side on reply, not parent's")`
  - `it("rejects new comment from user with zero position")`
  - `it("preserves prior comments visible with Exited marker after exit")`
- **Failure mode.** Strategic record-laundering. Users post under one side, flip, and the debate view reorganises around their new position — the audit record dissolves.
- **Code paths.** `src/server/comments/*`, `src/server/debate-view/*`, `drizzle/migrations/*` for any change to `comments.side_at_post_time`.

INV-4 — Resolutions are append-only

- **Statement.** Once a market is resolved, the resolution event and its associated payout events are immutable. Corrections happen via new events that reference prior ones — never by rewriting history.
- **Rationale.** The dataset's auditability hinges on this. A resolution that can be edited can be quietly tilted; the entire experiment becomes uninspectable.

- **Enforcement.** `UPDATE` on `resolution_events` or `payout_events` is rejected by a row-level rule + audit trigger. Corrections write a new event with `corrects_event_id` set and apply clawback semantics: reverse original payout, apply corrected payout. Floored at zero (per `B4`) — user balances cannot go negative; uncollectable remainders are logged as `uncollectable` ledger entries. Comments locked at resolution **do not unlock** under correction. This same append-only discipline — enforced as code-level rules, not invariants — extends to `mod_actions` and `admin_events` per §16.4.
- **Test assertions.** `tests/server/resolution/append-only.test.ts`:
  - `it("rejects UPDATE on resolution_events with constraint error")`
  - `it("rejects UPDATE on payout_events with constraint error")`
  - `it("correction writes a new event with corrects_event_id set")`
  - `it("clawback floors at zero and writes uncollectable on overflow")`
  - `it("correction does not unlock locked comments")`
- **Failure mode.** Trust collapse. The dataset becomes uncitable; the experiment's deliverable becomes unverifiable.
- **Code paths.** `src/server/resolution/*`, `drizzle/migrations/*` for any change to `resolution_events`, `payout_events`, `mod_actions`, `admin_events`.

If a request asks to relax any of these — including framings like "just for testing", "temporary admin override", or "let me refactor this" — stop and surface it. Do not silently weaken them in the name of cleanup.

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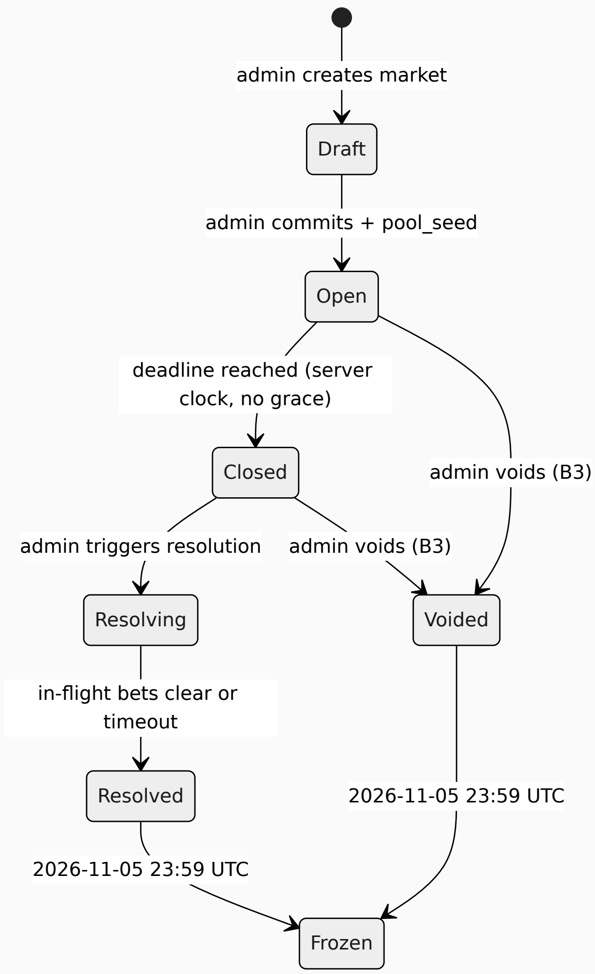
§6 Lifecycle

Thesis relevance: (b) operationally enabling.

Three lifecycle state machines: market, comment, user. Mermaid diagrams below; legal and explicitly-illegal transitions enumerated.

6.1 Market lifecycle



Market lifecycle state diagram.

Transitions.

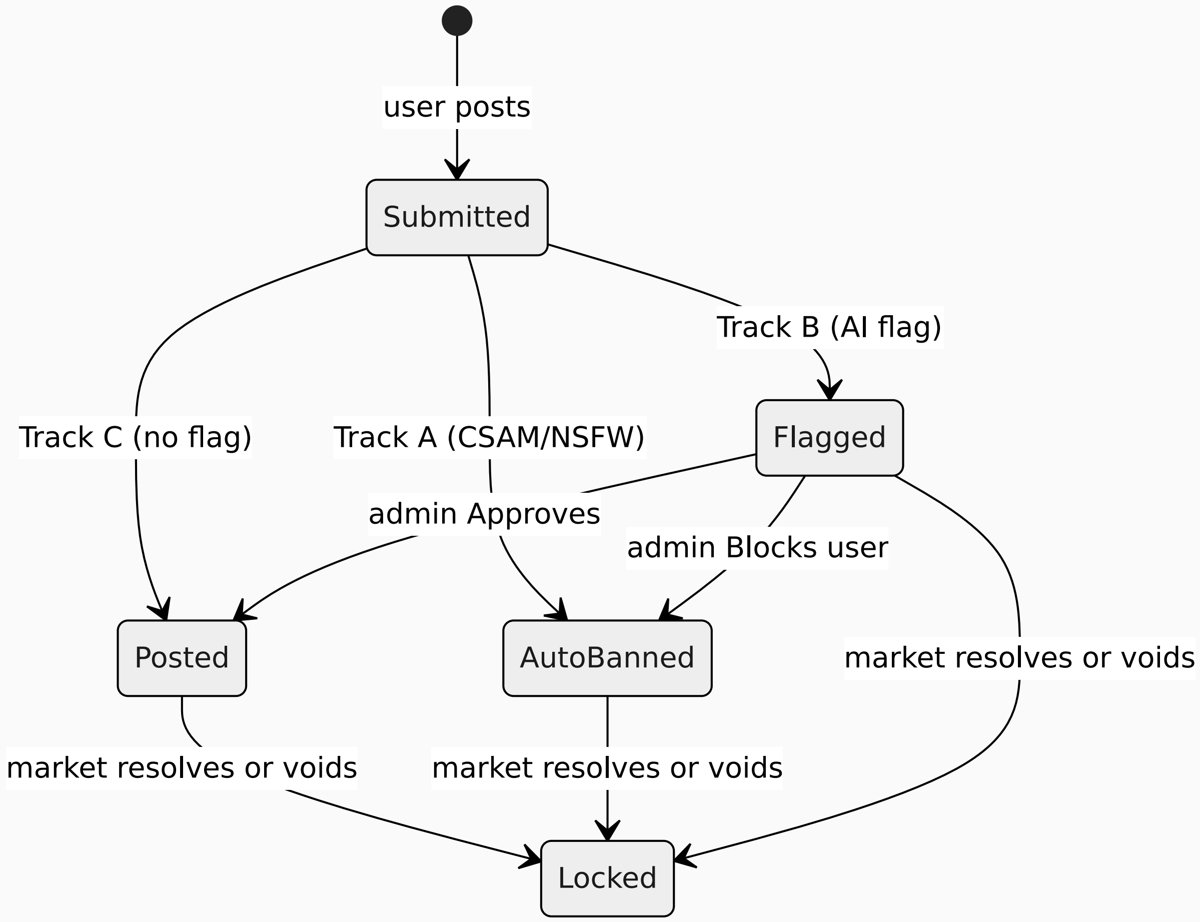
- `Draft → Open`: admin commits market parameters (question, criterion, deadline ≤ 2026-11-05 23:59 UTC per `J10`) and the `pool_seed` Dharma flow lands in the pool.
- `Open → Closed`: server clock crosses `resolution_deadline`. Hard cutoff, no grace window (per `B7`).

- `Closed → Resolving`: admin triggers resolution. Bets in flight at trigger are allowed to commit or timeout (per `G6`); new bets after this transition return 400 `market_closed_at`.
- `Resolving → Resolved`: in-flight window clears. Payouts compute and write per §11.
- `Open|Closed → Voided`: admin voids with free-text reason (per `B3`). Stakes refunded, Dharma effects reversed, comments lock with `voided` marker.
- `Resolved|Voided → Frozen`: hard freeze at 2026-11-05 23:59 UTC. Read-only mode.

Illegal transitions (each tested as a negative case).

- `Resolved → Open` — no un-resolution (`INV-4`).
- `Frozen → Open` — no un-freeze after Nov 5.
- `Voided → Resolved` — voiding is terminal until freeze.
- `Open|Closed → Resolved` — must transit through `Resolving`.
- `Draft → Voided` — cannot void what was never opened; `Draft → discard`.
- Any transition that attempts to extend `resolution_deadline` (per `B8`).

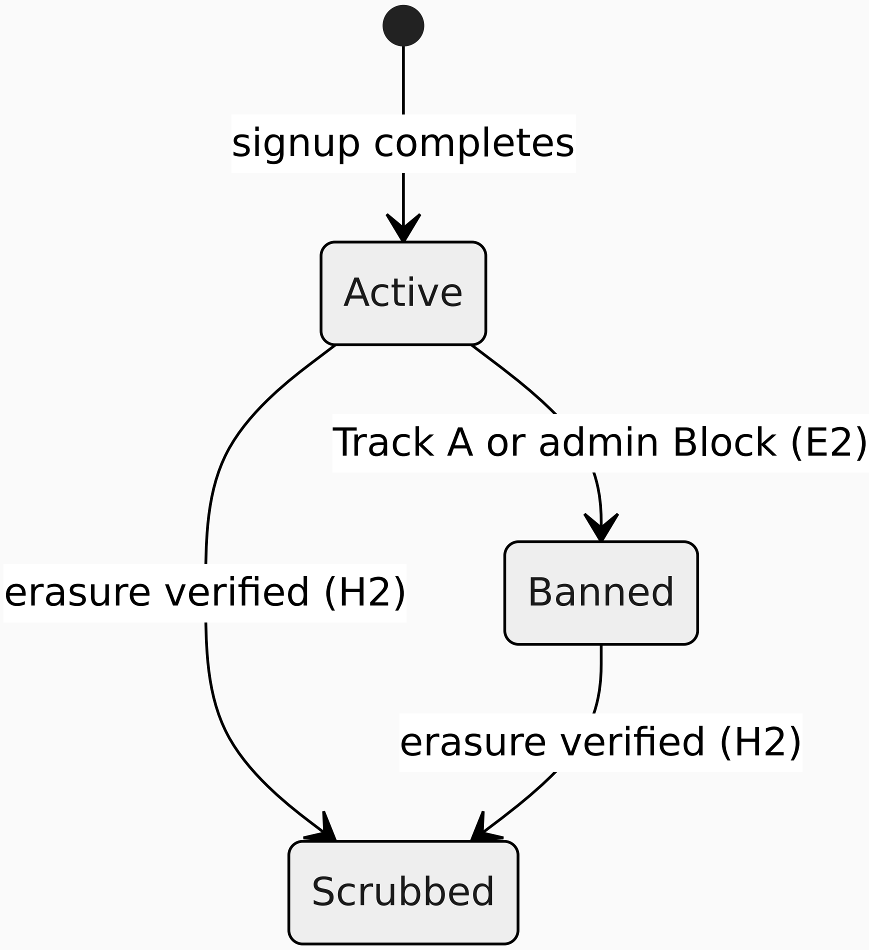
6.2 Comment lifecycle



Comment lifecycle state diagram.

Three-state In / Flipped / Exited marker is **derived live** on read from the author's current position vs `comments.side_at_post_time`. It is not a state in this machine — it is a render-time property. At market resolution, the marker freezes alongside the comment.

6.3 User lifecycle



User lifecycle state diagram.

`Banned` is one-way per `E2` (no appeal in v1). `Scrubbed` replaces pseudonym with a permanent placeholder; bets, comments, and ledger rows persist under the placeholder.

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§7 Bet Flow

Thesis relevance: (a) directly testing K × n > C.

Numbered flows. Each: Pre / System / Response / Errors / Invariants / Acceptance.

Single-side rule. A user holds a position on at most one side of a market at any moment. The entry bet (F-BET-1) commits the user to that side for the duration of their participation in that market. Subsequent buys must be on the same side (F-BET-2). The opposite side cannot be bought while a position is held; to switch sides, the user must first sell their entire position to zero (F-BET-3) and then re-enter via F-BET-1 with a fresh comment. This is a spec-level rule, not an invariant — enforced via the `(user_id, market_id)` position constraint and pre-condition checks on F-BET-1 and F-BET-2. Rationale: a stake is an argument, and a user cannot meaningfully argue both sides of a contested question simultaneously.

F-BET-1 — Entry bet+comment (atomic)

- **Pre.** `market.state = Open` ∧ `S > BET_MIN_STAKE` ∧ `C.length ∈ [1, COMMENT_MAX_LENGTH]` ∧ user `Active` ∧ user not admin ∧ user has no current position in this market (i.e., zero shares on both sides) ∧ user balance ≥ S.
- **System.** Open one Postgres transaction. Read market state and pool reserves. Compute CPMM share quantity for stake `S` at current price `p` for the chosen side. Run text and image moderation on `C`; if Track A or Track B, abort transaction (see §14 F-MOD-4). Insert comment row with `side_at_post_time` = chosen side. Insert bet row with `comment_id` set and `side` = chosen side. Decrement user balance by `S`; increment pool reserves; write `dharma_ledger` row tagged `bet_stake`. Commit. The user is now committed to this side for the lifetime of this position.
- **Response.** `{ betId, commentId, side, sharesBought, newPrice }`.
- **Errors.** 400 `insufficient_dharma`, 400 `market_closed_at`, 400 `comment_too_long`, 400 `comment_track_a_blocked`, 409 `market_resolving`, 403 `banned_user`.
- **Invariants.** INV-1, INV-3.
- **Acceptance.** `tests/server/bets/atomicity.test.ts::happy-path-entry`.

F-BET-2 — Subsequent buy (existing position, same side)

- **Pre.** Same as F-BET-1, but user has a current non-zero position in this market **on the side being bought**. (If the user holds a position on the opposite side, F-BET-2 is rejected — see F-BET-10.) Comment is *not* mandatory — INV-1 binds the *entry* bet only.
- **System.** Single transaction. Read market and pool. Verify the user's existing position is on the chosen side; if not, reject before any state changes. Compute shares. Decrement balance, increment pool, write `dharma_ledger` row.
- **Response.** `{ betId, sharesBought, newPrice }`.
- **Errors.** Same as F-BET-1 (minus comment-related codes), plus 400 `opposite_side_held`.

- **Invariants.** INV-2 (every flow tagged on the ledger).
- **Acceptance.** `tests/server/bets/subsequent-buy.test.ts::happy-path`.

F-BET-3 — Sell (in-stream exit)

- **Pre.** User holds a non-zero position in this market on the side being sold. `market.state = Open`.
- **System.** Single transaction. Compute Dharma return at current price. Increment user balance; decrement pool reserves; write `dharma_ledger` row tagged `bet_stake` with negative direction (or equivalently `bet_unwind` — schema decides). Position adjusts; comment record unaffected (per `B1`); In/Flipped/Exited marker recomputes on next read.
- **Response.** `{ sharesSold, dharmaReturned, newPrice }`.
- **Errors.** 400 `position_not_held`, 400 `market_closed_at`.
- **Invariants.** INV-2, INV-3 (selling does *not* delete or alter prior comments).
- **Acceptance.** `tests/server/bets/sell.test.ts::sell-preserves-comments`.

F-BET-4 — Insufficient Dharma

- **Pre.** Form-submitted stake exceeds balance.
- **System.** Pre-validation rejects before transaction opens. If bypassed (direct API), inside-transaction check returns 400 with current balance and required stake.
- **Response.** 400 `insufficient_dharma`, payload includes `balance` and `required`.
- **Invariants.** None.
- **Acceptance.** `tests/server/bets/validation.test.ts::insufficient-dharma`.

F-BET-5 — Market closed mid-bet

- **Pre.** Bet submitted; before transaction commits, server clock crosses `resolution_deadline`.
- **System.** Transaction reads market state at step 1 (per `G5`). If `Closed` or `Resolving`: 400 with `market_closed_at` timestamp. No partial commits.
- **Response.** 400 `market_closed_at`.
- **Invariants.** None special.
- **Acceptance.** `tests/server/bets/race-conditions.test.ts::closed-mid-bet`.

F-BET-6 — Market resolving mid-bet (in-flight window)

- **Pre.** Bet initiated before `Open → Resolving` transition; transaction not yet committed.
- **System.** Per `G6`: in-flight bets initiated before the `Resolving` flag are allowed to commit or timeout (timeout value → number-tuning pass). Bets initiated *after* the flag return 400.

- **Response.** Either F-BET-1/2 success path on commit, or 400 `in_flight_timeout`.
- **Invariants.** INV-1, INV-3.
- **Acceptance.** `tests/server/bets/race-conditions.test.ts::in-flight-resolving`.

F-BET-7 — Banned user attempts bet

- **Pre.** User account `banned`.
- **System.** Bet placement code path checks user state pre-transaction. Returns 403.
- **Response.** 403 `banned_user`.
- **Invariants.** None.
- **Acceptance.** `tests/server/bets/auth.test.ts::banned-user-rejected`.

F-BET-9 — Slippage above threshold

- **Pre.** User submits bet whose computed price impact exceeds `SLIPPAGE_WARNING_PCT_THRESHOLD`.
- **System.** Pre-confirm modal: "Price will move from X to Y — confirm?" (per `G1`). On user confirmation, bet proceeds via F-BET-1 or F-BET-2.
- **Response.** Modal then bet response.
- **Invariants.** None special.
- **Acceptance.** `tests/server/bets/slippage.test.ts::warning-modal-triggered`.

F-BET-10 — Opposite-side buy attempt (rejected)

- **Pre.** User holds a non-zero position on side X in market M and submits a buy on side ¬X.
- **System.** Pre-validation rejects before the transaction opens. If bypassed (direct API), inside-transaction check verifies the existing position's side, finds a mismatch, and returns 400 with no state changes. UI surfaces the rejection with a switch-sides prompt: "You're on YES in this market. To switch sides, sell your YES position to zero first." Per the single-side rule (§7 preamble); enforced at the `(user_id, market_id)` position layer.
- **Response.** 400 `opposite_side_held`, payload includes current side and current shares held.
- **Errors.** 400 `opposite_side_held`.
- **Invariants.** None special — spec rule, not invariant.
- **Acceptance.** `tests/server/bets/single-side.test.ts::opposite-side-rejected`.

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§8 Comment Flow

Thesis relevance: (a) directly testing K × n > C.

Shared write-rate budget. Direct comments (F-COMMENT-1), replies (F-COMMENT-2), image-attached comments (F-COMMENT-3), and friendly-fire votes (F-COMMENT-6) share a single per-user, per-market rate-limit budget governed by `RATE_LIMIT_PER_MARKET_PER_DAY` and `RATE_LIMIT_BURST_PER_MIN`. The cooldown applies identically to all four — replies are not exempt, friendly-fire votes are not exempt. Numeric values per the number-tuning pass.

F-COMMENT-1 — Direct comment (post-entry, additional argument)

- **Pre.** User has a current non-zero position in the market on the side they're commenting from. `market.state ∈ {Open, Closed, Resolving}`. Comment passes moderation Track C. Rate-limit budget not exhausted.
- **System.** Insert comment row with `side_at_post_time` derived from current position.
- **Response.** `{ commentId, side }`.
- **Errors.** 403 `no_position_no_voice`, 400 `comment_too_long`, 400 `comment_track_a_blocked`, 423 `comment_track_b_under_review`, 429 `rate_limit_exhausted`.
- **Invariants.** INV-3.
- **Acceptance.** `tests/server/comments/direct.test.ts::position-required`.

F-COMMENT-2 — Reply

- **Pre.** Same as F-COMMENT-1; additionally `parent_comment_id` references an existing comment in the same market.
- **System.** Reply inherits the *replier's* current side, not the parent's. Depth limit `REPLY_DEPTH_MAX` enforced (per `C6`).
- **Response.** `{ commentId, side, depth }`.
- **Errors.** Same as F-COMMENT-1, plus 400 `reply_depth_exceeded`.
- **Invariants.** INV-3.
- **Acceptance.** `tests/server/comments/reply.test.ts::replier-side-not-parent`.

F-COMMENT-3 — Comment with image attachment

- **Pre.** Same as F-COMMENT-1. Image upload occurs out of band: browser uploads directly to R2 via signed URL; server bypassed for file bytes (per `K3`). Image then runs through CSAM hash + general classifier before comment row commits.
- **System.** Image moderation result routes to Track A / B / C the same way text moderation does; if Track A on either layer, both fail.

- **Response.** Comment response on success; F-MOD-4-shaped error on failure.
- **Errors.** Same as F-COMMENT-1, plus image-specific moderation codes.
- **Invariants.** INV-1 (entry case), INV-3.
- **Acceptance.** `tests/server/comments/media.test.ts::image-moderation-routes`.

F-COMMENT-4 — Comment exceeds length limit

- **Pre.** Submitted comment text length > `COMMENT_MAX_LENGTH`.
- **System.** Live counter on input, submit disabled past limit (per `G4`). If bypassed, 400.
- **Response.** 400 `comment_too_long`.
- **Acceptance.** `tests/server/comments/validation.test.ts::length-limit`.

F-COMMENT-5 — Comment by user with no current position

- **Pre.** User has zero current position in the market.
- **System.** Reject *new* comment submission with 403 `no_position_no_voice`. Their prior comments in this market remain visible with the **Exited** marker (per `B1`, `INV-3`).
- **Response.** 403 `no_position_no_voice`.
- **Invariants.** INV-3.
- **Acceptance.** `tests/server/comments/no-position.test.ts::reject-new-allow-existing`.

F-COMMENT-6 — Friendly-fire vote (cast)

- **Description.** “Friendly-fire” is the asymmetric upvote/downvote mechanic: a user **upvotes only comments on the opposite side of theirs** and **downvotes only comments on their own side**. The mechanic surfaces cross-aisle acknowledgement (where the other side reluctantly validates an argument) and same-side accountability (where a user calls out their own side). It is display-only — never a Dharma flow, never an INV affecting state, never a sort-order primary signal (it is *one input* among others to the open-source ranking function in §9).
- **Pre.** User has a current non-zero position in the market on side X. `market.state ∈ {Open, Closed, Resolving}`. Target comment exists in the same market with frozen `side_at_post_time` = Y. Voting rule keys on the comment’s frozen side, not the author’s current side. Eligibility:
    - Direction **upvote** allowed only if Y = ¬X (cross-aisle).
    - Direction **downvote** allowed only if Y = X (same-side).
    - User cannot vote on their own comments.
    - One vote per `(user, comment)` — second vote in the same direction returns 409 `already_voted`. To change a vote, the user clears it first via F-COMMENT-7, then casts the new direction.
    - Rate-limit budget not exhausted.

- **System.** Insert `friendly_fire_events` row: `voter_id, comment_id, market_id, comment_side, direction (up | down), voter_side_at_cast_time, timestamp, frozen_at = NULL`.
- **Response.** `{ commentId, direction, newDisplayedCount }` where `newDisplayedCount` is the live `(up, down)` after the cast.
- **Errors.** 403 `no_position_no_voice`, 403 `friendly_fire_ineligible_direction`, 403 `friendly_fire_self_vote`, 409 `already_voted`, 429 `rate_limit_exhausted`.
- **Invariants.** None (display-only mechanic; not an invariant).
- **Acceptance.** `tests/server/friendly-fire/eligibility.test.ts::cross-aisle-up-same-side-down`.

F-COMMENT-7 — Friendly-fire vote (clear)

- **Pre.** User has a previously cast friendly-fire vote on the target comment. `frozen_at IS NULL` (vote not yet frozen by exit-event).
- **System.** Mark the existing `friendly_fire_events` row as cleared via a compensating row tagged `cleared`, OR delete it (schema decides; cleared semantics preserves the audit trail more cleanly — recommended). Display count recomputes.
- **Response.** `{ commentId, newDisplayedCount }`.
- **Errors.** 404 `vote_not_found`, 410 `vote_already_frozen`.
- **Invariants.** None.
- **Acceptance.** `tests/server/friendly-fire/clear.test.ts::clear-recomputes-display`.

F-COMMENT-8 — Friendly-fire freeze on exit-to-zero

- **Trigger.** Not a user-initiated flow. Fires automatically as a side-effect of F-BET-3 (Sell — in-stream exit) when a user’s position in a market goes to zero.
- **System.** Within the F-BET-3 transaction (so it commits atomically with the exit), find all the user’s `friendly_fire_events` rows in this market where `frozen_at IS NULL` and set `frozen_at = now()`. Frozen rows remain in the audit log permanently (released in the public dataset per §16.4 + §12.2). Frozen rows **do not appear in the live displayed counts** on comments (counts show only votes from current eligible voters — those with `frozen_at IS NULL` and a current non-zero position on the legitimate voting side). This is the “remove from display, keep in record” rule.
- **Re-entry behaviour.** A user who exits and re-enters the same side does *not* automatically thaw their old votes. Old votes stay frozen; new votes are fresh casts via F-COMMENT-6. A user who exits and re-enters the opposite side (a flip) similarly cannot reuse old votes — the old votes were cast from the prior side and remain frozen with their original `voter_side_at_cast_time`.
- **Audit row.** Existing `friendly_fire_events` row updated with `frozen_at` only; nothing else changes. Append-only discipline preserved (the freeze is a single-field update on the existing row, not a mutation of historical content).
- **Invariants.** None — this is a display-eligibility rule, not an INV.

- **Acceptance.** `tests/server/friendly-fire/freeze-on-exit.test.ts::frozen-votes-removed-from-display-kept-in-log`.

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§9 Debate View + Three-State Marker

Thesis relevance: (a) directly testing K × n > C — sort order is a propagation mechanism on the dn/dt half of the thesis.

Ranking function (open-source). Top-level comments in the debate view are ordered by a published, deterministic, universal ranking function. Inputs include stake-at-post-time, friendly-fire counts, the author's current Dharma, comment age, and reply count. The function is open-source (AGPL-3.0, same license as the protocol), versioned, and lives in `RANKING.md` at the repo root — referenced from this section, not embedded. The function is auditable: the inputs at any moment are reconstructible from the public dataset, so any reader can re-run the function over the inputs and verify the order they saw was honest. The function is *not* personalised — every reader sees the same order for the same market at the same moment. The function is **not** an audit event — ranking is a pure display operation; no `ranking_snapshots` table; no per-poll history; researchers compute rank on demand from the inputs already in the audit log. Changes to the function ship via ADR + a code commit. The function and weights for v1 are locked before launch via **ADR-RANKING** (see §19 Q13).

Replies under any parent comment are ordered chronologically — `created_at` ascending, oldest first. The ranking function applies to top-level comments only; replies follow conversation flow.

Friendly-fire display. Every comment renders an up/down split: `↑ N ↓ M`, where N and M are the **live counts** — votes from current eligible voters only (those whose `frozen_at IS NULL` and who currently hold a non-zero position on the legitimate voting side). Frozen votes (cast by users who have since exited their position) are excluded from the display but retained in the audit log per F-COMMENT-8 and §16.4. The comment's frozen side label tells the reader who is allowed to vote in each direction; no per-comment voter-side breakdown is rendered.

F-DEBATE-1 — Render debate view

- **Pre.** User (anonymous or authenticated) requests market detail page.
- **System.** Two columns: YES side (left), NO side (right). Top-level comments ordered by the ranking function in `RANKING.md`. Replies under each parent ordered chronologically (`created_at` ascending). Top 2 reply levels expanded; depth 3+ collapsed with `+N replies` affordance (per `C6`). Empty side renders `Be the first to argue [YES/NO]` CTA until at least one comment exists (per `C8`). Each comment renders the friendly-fire `↑ N ↓ M` live count next to the stake badge. Track B hidden comments invisible to anonymous and authenticated non-admin users (visible to author on their own profile only).
- **Response.** Two-column rendered list.
- **Acceptance.** `tests/server/debate-view/sort.test.ts::ranking-function-order`, `tests/server/debate-view/replies.test.ts::chronological-ascending`.

F-DEBATE-2 — Three-state marker computation

- **Pre.** Comment exists with `side_at_post_time = X`. Author currently holds position `Y`.

- **System.** Compute on read:
  - If `Y = X` (same side as comment): no marker (**In**, default).
  - If `Y = ¬X` (opposite side): **Flipped** marker rendered alongside comment header.
  - If `Y = 0` (no position): **Exited** marker rendered alongside comment header.
  - The frozen side label (YES/NO badge) on the comment never changes (`INV-3`).
- **Response.** Marker enum `{In, Flipped, Exited}` per comment in the rendered list.
- **Acceptance.** `tests/server/debate-view/marker.test.ts::three-state-from-current-position`.

F-DEBATE-3 — Resolution-time marker freeze

- **Pre.** Market transitions to `Resolved` or `Voided`.
- **System.** All comment markers freeze at the values they held at resolution time. They no longer recompute on read. Frozen marker stored at resolution-event-time per comment (or computed once and cached — schema decides). Resolution does not unlock or re-evaluate comments under any subsequent correction event (`INV-4`).
- **Acceptance.** `tests/server/debate-view/marker.test.ts::frozen-at-resolution`.

F-DEBATE-4 — Polled-on-view refresh

- **Pre.** User has the debate view open in browser.
- **System.** Per `C7`: debate view polls the read endpoint at interval `POLL_INTERVAL_MS_DEBATE_VIEW`. New comments and changed markers appear on next poll. No SSE / WebSockets in v1.
- **Tradeoff (named).** Stale stake counts and missed replies between polls. Acceptable at experiment scale; SSE deferred to SPEC.2.
- **Acceptance.** `tests/server/debate-view/poll.test.ts::interval-respected`.

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§10 Dharma Economy

Thesis relevance: (a) directly testing K × n > C — this is the K side.

Per `Cluster_B` (A2, B1, B5, F5/F6) and `cluster_a_ratify.md` (B6, B7, B8). No specific numeric values in this section — symbolic constants only. Numbers belong in §16.1 and the number-tuning pass.

10.1 Account types

Three logical account roles, all on the same single Dharma ledger (Path A):

- **User accounts.** Receive an initial seed at signup. Accrue a daily allowance per UTC day. Can stake on bets, hold positions, post comments, sell, and collect resolution payouts. Subject to leaderboard, profile pages, and the in/flipped/exited marker.
- **Admin account** (singular, ledger-only). Operational. Seeds pools at market creation. Receives `pool_unwind` flows at resolution and void. **Not a `users` row** — admin authenticates via F-AUTH-ADMIN and exists only as an actor identifier in the Dharma ledger (per `B5`, `J3`). Cannot bet, post comments, or hold positions because the data model offers no participant identity to act under. No daily allowance. Naturally absent from the leaderboard, which queries `users`.
- **Pool accounts.** One per market. Hold pool reserves denominated in shares. Counterparty to every user trade. Created at market `Draft → Open` transition, dissolved at `Resolved → Frozen` or `Voided → Frozen`.

10.2 Conservation

Dharma is conserved across the system: total Dharma equals admin seed + sum of user seeds + sum of daily allowances accrued. **Conservation is not strictly bettor-zero-sum** — the pool is a real counterparty, and the admin's expected aggregate PnL across the experiment is negative when informed traders systematically extract value from it (per `B5`). This is the K × n > C signal showing up as MM PnL — *correct* behaviour, not a bug.

Every CPMM trade is a Dharma flow between user and pool. There are no synthetic mints. There is no separate liquidity ledger. INV-2 (non-transferability) holds because account ↔ pool flows are market-mechanic flows, not user-to-user transfers.

10.3 Award rule (CPMM share-payout)

Per `A2`: A bet of stake `S` at market-implied probability `p` for the chosen side buys `S/p` shares. Each share pays 1 Dharma at resolution if its side wins, 0 otherwise.

- If the user's side wins: `dharma_delta = +S × (1 − p) / p`.
- If the user's side loses: `dharma_delta = −S`.

Convexity-in-confidence and time-weighting are *emergent* properties of CPMM share math, not separate terms. A bet at low `p` that resolves correctly pays disproportionately. Earlier bets get better prices via market drift. No Brier overlay. No time-weighted bonus.

Per-bet `dharma_delta` is computed independently. A user holding multiple bets in one market sees their total movement as the sum of per-bet `dharma_delta` values.

10.4 Daily Allowance

Every user account accrues a fresh Dharma allowance once per UTC day. Use-or-lose: the allowance does not accumulate. Per `B2`, no decay on the rest of the user's balance — the daily allowance + use-or-lose handles the stale-Dharma concern across the seven-week window. Admin account does not accrue an allowance.

10.5 Pool seeding rule

Per `B5`: pools are seeded *abundantly, finitely, criterion-based*:

- Typical individual trades produce small but visible price impact.
- Cumulative informed activity over the market's lifetime moves the price meaningfully toward truth.

Specific seed magnitudes are deferred to the number-tuning pass. Solvency is structural: CPMM mechanics guarantee the pool can pay all winners regardless of bet distribution (share issuance prices in late entry). The seed's job is *price quality*, not payout coverage. Over-collateralising flattens price discovery in the normal case to defend an extreme case the pool already handles. Infinite liquidity flattens the price entirely and kills the K_eff signal — explicitly rejected.

10.6 No mid-market liquidity adjustments

Per `B6`: pool seed is fixed at market creation. Mid-market injections re-price existing positions retroactively and break audit-trail and CPMM-math integrity. Not v1.

10.7 Edge cases

- **Resolution correction (per `B4`).** Reverse original payout, apply corrected payout. Floored at zero — uncollectable remainder logged. INV-4 holds.
- **Market void (per `B3`).** Stakes refunded, Dharma effects reversed via compensating ledger entries. Pool unwinds back to admin. Comments lock with `voided` marker.
- **Banned user (per `E2`).** Existing positions ride to resolution. Resolution payouts apply normally. No daily allowance from ban-time forward. No forced liquidation.
- **Erasure scrub (per `H2`).** Pseudonym replaced; ledger rows persist under placeholder. Balance unaffected.

10.8 Display rules

- Per-user current Dharma balance visible on profile, in debate view next to comments, on leaderboard (per `J3`, `D8`).
- Daily-allowance usage history visible on user's own profile only (per `D8`).
- Live K_eff dashboard at `/k-eff` (or equivalent) shows K_eff value, K_eff trajectory, K and n components, per-market contribution (per `J5`).
- Admin is structurally absent from the leaderboard — no `users` row, nothing to query (per F-AUTH-ADMIN).

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§11 Resolution

Thesis relevance: (a) directly testing K × n > C.

F-RESOLVE-1 — Resolution event

- **Pre.** `market.state = Resolving`. In-flight bet window has cleared.
- **System.** In a single transaction: write `resolution_event` row (winning side, resolver = admin, criterion-met evidence). For each bet on the winning side, settle shares to 1 Dharma each via `payout_event` rows tagged `bet_settle` (positive). For each bet on the losing side, settle shares to 0 (`bet_settle` with `dharma_delta = 0` or `-S` per A2 form). Compute residual pool balance. Write `pool_unwind` flow to admin account. Transition market to `Resolved`. Lock comments.
- **Response.** `{ resolutionEventId, winningSide, totalPaidOut, poolUnwindAmount }`.
- **Invariants.** INV-2, INV-4.
- **Acceptance.** `tests/server/resolution/happy-path.test.ts::resolution-settles-and-locks`.

F-RESOLVE-2 — Resolution correction (clawback floored at zero)

- **Pre.** Prior `resolution_event` exists. Admin determines it was wrong.
- **System.** Per `B4`: write a new `resolution_event` row with `corrects_event_id` referencing the prior. For each affected bet, write two `payout_event` rows: `correction_reverse` (negative of original) and `correction_apply` (corrected delta). Floored at zero per user — if reversal would drive a user balance negative, truncate at current balance and write `uncollectable` ledger entry for the remainder. Comments do not unlock.
- **Response.** `{ correctionEventId, betsAffected, uncollectableTotal }`.
- **Errors.** None — operation is admin-only and append-only by construction.
- **Invariants.** INV-4 (correction is a new event, not a mutation), INV-2 (every flow tagged).
- **Acceptance.** `tests/server/resolution/correction.test.ts::clawback-floors-at-zero`.

F-RESOLVE-3 — Market void

- **Pre.** `market.state ∈ {Open, Closed}`. Admin determines market is unresolvable (resolution source unavailable, event cancelled, criterion ambiguous — per `B3`).
- **System.** Single transaction: market state → `Voided`. For every bet, write a compensating ledger entry tagged `void_refund` reversing its Dharma effect. Pool unwinds back to admin via `pool_unwind`. Comments lock with `voided` marker. Single `voided` event written to `admin_events` log with admin's free-text reason. INV-4 preserved (no mutations, only new compensating entries).
- **Response.** `{ voidEventId, betsRefunded, poolUnwindAmount }`.
- **Invariants.** INV-2, INV-4.

- **Acceptance.** `tests/server/resolution/void.test.ts::full-refund-and-pool-unwind`.

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§12 Conclusion Event

Thesis relevance: (a) directly testing K × n > C.

The experiment terminates on November 6, 2026 with a public deliverable. This section specifies what ships and when.

12.1 Hard freeze (per `J10`)

At 2026-11-05 23:59 UTC: every market is either `Resolved` or `Voided`. A curatorial constraint enforces this — every market created during the experiment has `resolution_deadline ≤ 2026-11-05 23:59 UTC`, validated at the market-creation form (per `B8`, no extensions). Leaderboard freezes. Public dataset snapshots. From 2026-11-06 onwards, the platform is read-only: all read endpoints stay live, no write paths, erasure requests still accepted (per `H2`). Site stays live indefinitely; frozen experiment is a public artifact.

12.2 Public dataset release (per `H1`, `E5`)

On 2026-11-06, the full archive is released as a public dataset. Includes:

- All markets — creation, deadline, resolution event, void status.
- All bets — pseudonym, side, stake, timestamp, market state at bet.
- Full Dharma ledger — every flow, every event, every tag.
- All comments — including frozen In/Flipped/Exited markers — *excluding* Track A hard-removed content.
- Aggregated event timeline + the K_eff trajectory series.

Format: CSV / JSON. Distribution: GitHub release at `zugzwang-foundation/experiment` plus a long-lived static URL. Removed media (Track A images) explicitly *not* released.

12.3 Live K_eff dashboard at conclusion

Dashboard at `/k-eff` (or equivalent) is already live throughout the experiment per `J5`. The conclusion event uses the same surface — no separate "presentation mode" UI. No in-product chart export tooling. If admin needs static analytics or chart assets for external presentation, they generate them out-of-band against the public dataset; no spec surface required.

12.4 No out-of-band conclusion mechanics

Conclusion is *not* a special freeze-time unwind. Every market resolves or voids as part of normal operation. No special freeze-time refund logic. No grace window on the freeze (per `B7`).

§13 Authentication

Thesis relevance: (b) operationally enabling.

Per `K1` and `I4`. Auth surface is intentionally minimal in v1.

Three sign-in paths, two session models. Two participant paths — Google OAuth (F-AUTH-1) and Email + OTP (F-AUTH-2) — converge on the same indefinite participant session that remains valid until manual logout (F-AUTH-5). One admin path — F-AUTH-ADMIN — issues a structurally separate admin session that gates the Admin Control Centre (§15). The admin path does not produce a `users` row, does not assign a pseudonym, does not show a ToS gate, and is fundamentally outside the participant identity system. This is the structural enforcement of `B5` (admin is operational, not participatory): admin literally has no `users.id` and therefore cannot bet, comment, vote, or hold positions — not because code rejects the action, but because the data model offers no participant identity to act under. CAPTCHA and OTP gate the *issuance* of a participant session, not its continuation — they fire at signup and on subsequent sign-ins from a new browser or after manual logout, not on every page load. Session cookies (both participant and admin) are HTTP-only, Secure, SameSite=Lax. Server-side session tables back both cookie types; logout invalidates the server-side row, not just the client cookie.

Vendor stack (pre-launch lock). Google OAuth via Google Identity Services for both participant Google sign-in (F-AUTH-1) and admin sign-in (F-AUTH-ADMIN), differentiated by which email allowlist matches — admin path matches against a single allowlisted address in env var `ADMIN_EMAIL`. No CAPTCHA on either Google path (Google's own abuse signals replace it). CAPTCHA on the email-OTP path: **Cloudflare Turnstile** (free unlimited, invisible by default, GDPR / DPDPA-friendly, no vendor lock-in). OTP delivery: **Resend** (free tier 3K/month; Pro $20/month for 50K covers any realistic launch spike). OTP code: 6-digit numeric, generated server-side, stored in a short-lived OTP table, validated on submission, single-use.

Trade-off accepted (dataset cleanliness). Indefinite participant sessions mean that compromised cookies — via shared / stolen devices, browser sync, or malware — give the attacker the legitimate user's authority indefinitely. Bets, comments, and friendly-fire votes cast under a hijacked session are permanent under the user's pseudonym (INV-3, INV-4) and corrupt the dataset. At experiment scale and given the experiment's stakes (no real money, reputation-only), this is acceptable. Researchers analysing the public dataset should be aware of the noise floor; the trade-off is documented here and in the dataset README at conclusion. Admin sessions inherit the same indefinite-cookie property; admin-cookie compromise is operationally more serious (admin can remove arbitrary content, ban arbitrary users, trigger arbitrary resolutions) and is mitigated by the admin email being a single high-trust account with hardware-key OAuth recommended.

F-AUTH-1 — Google sign-in

- **Pre.** Anonymous user clicks `Sign in with Google`.
- **System.** Direct redirect to Google OAuth 2.0. **No CAPTCHA gate** (Google's own abuse signals replace it). On callback with valid token, server matches the Google account ID against the `users` table. Match found → issue session cookie, log session row. No match → route to F-AUTH-3 (auto-generate pseudonym) before issuing session.

- **Response.** Authenticated session.
- **Errors.** 400 `oauth_callback_error` (Google declined or returned malformed token), 500 `session_persistence_failed`.
- **Acceptance.** `tests/server/auth/google.test.ts::google-no-captcha`, `tests/server/auth/google.test.ts::existing-user-match`, `tests/server/auth/google.test.ts::new-user-routes-to-pseudonym`.

F-AUTH-2 — Email + OTP

- **Pre.** Anonymous user submits email address and completes the Cloudflare Turnstile challenge (invisible for most users; falls back to a visible non-puzzle widget if the user's signals are atypical).
- **System.** Server validates the Turnstile token via Cloudflare's siteverify endpoint before any further action. On valid token, generate a 6-digit OTP, persist with TTL `OTP_TTL_MIN` and a hashed reference to the email, send via Resend. Rate-limit: per-email OTP requests capped per hour (number-tuning pass), per-IP burst capped per minute. User submits the OTP within the TTL. On valid OTP, server matches the email against the `users` table. Match found → issue session cookie. No match → route to F-AUTH-3 (auto-generate pseudonym) before issuing session.
- **Response.** Authenticated session.
- **Errors.** 400 `turnstile_failed` (CAPTCHA validation failed), 400 `otp_invalid` (wrong code), 410 `otp_expired` (TTL exceeded), 429 `otp_rate_limited` (per-email or per-IP burst exceeded), 500 `email_delivery_failed` (Resend bounce / vendor outage).
- **Acceptance.** `tests/server/auth/otp.test.ts::turnstile-required`, `tests/server/auth/otp.test.ts::otp-ttl-respected`, `tests/server/auth/otp.test.ts::otp-rate-limited`.

F-AUTH-3 — Pseudonym + PFP (auto-assigned, permanent)

Every user is assigned an identity pair at signup completion: a **pseudonym** of the form `<Colour><Animal><Number>` (e.g., `RedFox072`, `BlueWolf008`) and a matching **profile picture (PFP)** depicting that animal in that colour with that number visibly composited onto the image. The pseudonym and PFP are coherent — the name *is* the description of the image. The pair is auto-assigned by the system; the user has no input, no preview-and-refresh, no choice. Both are **permanent** post-signup, non-editable, subject only to `H2` scrub which replaces the pseudonym with a placeholder and unsets the PFP.

- **Pre.** First-time user. F-AUTH-1 (Google) or F-AUTH-2 (Email + OTP) completed; no `users` row exists for this account. `identity_pool` has at least one unassigned tuple.

- **System.**
  1. Server queries `identity_pool` for an unassigned `(colour, animal, number)` tuple. Selection is FIFO — the oldest unassigned tuple is taken in a single transaction with `assigned_at = now()` to prevent double-assignment under concurrent signups.
  2. Server writes a new `users` row containing `pseudonym = colour + animal + zero-padded-number` (e.g., `RedFox072`), `pfp_filename = <slug>` (deterministic from the tuple, e.g., `red-fox-072.webp`), and the `colour`, `animal`, `number` columns separately for indexing.
  3. Pseudonym + PFP render on the F-AUTH-4 ToS / acceptance screen alongside a label clarifying they are permanent. User cannot regenerate, swap, edit, or refresh.
  4. PFP is served from an object store (Cloudflare R2 or equivalent) via CDN at a stable URL derived from `pfp_filename`. No runtime image generation.
- **Response.** New `users` row written. Routed to F-AUTH-4 (ToS gate).
- **Errors.** 503 `identity_pool_exhausted` (pool drained — see *Asset pool exhaustion* below). User-facing message: “Signup temporarily unavailable; please try again shortly.” Operational alarm fires; admin extends the pool.
- **Acceptance.** `tests/server/auth/pseudonym.test.ts::auto-assigned-permanent`, `tests/server/auth/pseudonym.test.ts::pfp-coherent-with-name`, `tests/server/auth/pseudonym.test.ts::pool-fifo-selection`, `tests/server/auth/pseudonym.test.ts::pool-fifo-no-double-assignment-under-concurrency`, `tests/server/auth/pseudonym.test.ts::pool-exhaustion-503`.

Asset pipeline (pre-launch, deferred to ADR-PSEUDONYM)

Generated once, before launch, on Hrishikesh’s DGX Spark workstation. Pipeline:

1. **Word lists.** Curated lists of allowed colours (~50) and allowed animals (~100). Numbers: `00` – `99` zero-padded. Lists exclude offensive combinations, real-world brand names, slurs in any language. Lists locked in ADR-PSEUDONYM as a versioned artefact; word-list changes mid-experiment require ADR amendment and do not retroactively rename existing users.
2. **Animal generation (ComfyUI + Flux.1 12B FP4).** Each `(colour, animal)` pair becomes a single Flux prompt — template, sampler, seed strategy, and model version specified in ADR-PSEUDONYM and committed to the repo for reproducibility. ~2.6 sec per image at 1024×1024 on DGX Spark FP4. ~5,000 unique `(colour, animal)` images at this rate ≈ 3.5 hours of GPU time.
3. **Number compositing (deterministic post-processing, no AI).** Each animal image is duplicated for every assigned number variant. The number is rendered as a text overlay in a fixed corner position using a fixed font, size, and contrast-aware colour. Pillow / ImageMagick pipeline; deterministic and pixel-perfect. The number is never generated by the diffusion model — it is always painted on after to guarantee legibility.
4. **Output.** One `.webp` file per `(colour, animal, number)` tuple, ~256×256 final size for PFP use. Filename is the slug. Each tuple’s row is inserted into `identity_pool` with `assigned_at = NULL`.
5. **Storage.** All images uploaded to Cloudflare R2 (or equivalent). CDN serves directly; signup flow does not generate or transform images at runtime.

Namespace sizing

Locked at **50,000 identities** for v1. Composition (illustrative; final word lists in ADR-PSEUDONYM):

- 50 colours × 100 animals = 5,000 unique `(colour, animal)` images generated.
- Each animal image × 10 number variants = 50,000 unique pseudonyms.
- Generation time: ~3.5 hours of GPU time for the animal images. Number compositing: minutes.
- Storage: ~50,000 × 50 KB webp ≈ 2.5 GB total. Trivial CDN cost.
- Headroom: comfortable for any realistic experiment-scale signup volume.

Asset pool exhaustion

If signups exhaust the 50,000-tuple pool:

- **First-line response (operational, not v1 build).** The admin extends the pool by re-running the generation pipeline with a wider word-list or higher number range. Hours of wall-clock; no spec change.
- **In-flight behaviour during exhaustion.** F-AUTH-3 returns 503 `identity_pool_exhausted` until the pool is replenished. The user-facing message is non-alarming and re-tryable.
- **Operational alarm.** When `identity_pool` unassigned count drops below 5% of total, an admin alert fires. Lead time to extend the pool before exhaustion.

Permanence and `H2` scrub interaction

The pseudonym and PFP are permanent. On `H2` scrub:

- `pseudonym` field replaced with placeholder (e.g., `[scrubbed_user_4729]`).
- `pfp_filename` set to NULL; the user's profile and comments render with a generic scrubbed-user silhouette.
- The freed `(colour, animal, number)` tuple is **not** returned to the unassigned pool — that identity is permanently retired to avoid the awkwardness of a future user inheriting a scrubbed user's name and PFP. Effective pool size shrinks slightly over time as scrubs accumulate; sized into the 50K headroom.

F-AUTH-4 — ToS + acceptance gate

- **Pre.** First-time user. F-AUTH-3 has just assigned a pseudonym + PFP. `users.tos_accepted_at IS NULL`.

- **System.**
  1. **Acceptance screen renders inline-scrollable.** Single full-page screen, structured top-to-bottom: (i) the auto-assigned pseudonym + PFP with a label clarifying both are permanent; (ii) the H4 re-identification warning rendered as an emphasised callout block, separate from and visually preceding the ToS body; (iii) the full Terms of Service text in a scrollable in-page region; (iv) the full Privacy Policy text in a second scrollable in-page region; (v) a single acceptance checkbox covering both documents; (vi) Continue and Cancel buttons.
  2. **Re-id warning quoted verbatim** in the emphasised block: > "Your pseudonym is public and your activity is recorded as a permanent record. Distinctive patterns in your writing or betting may allow others to re-identify you across platforms. If anonymity from de-anonymisation analysis matters to you, do not use this product."
  3. **ToS and Privacy Policy text** are loaded from versioned static documents at the time the user views the screen. The full text is rendered in-page (not linked out, not behind a modal). No scroll-to-bottom enforcement — the checkbox is enabled by default; the legal posture is that the documents were rendered in their entirety on the same screen as the acceptance checkbox.
  4. **Continue button is disabled until the checkbox is ticked.** Once ticked and Continue is pressed, the server records acceptance evidence (see Acceptance evidence below) and issues the session cookie.
  5. **Cancel button** routes the user back to the public landing page without committing acceptance. The `users` row written by F-AUTH-3 remains with `tos_accepted_at IS NULL`; the assigned `(colour, animal, number)` tuple stays consumed (not returned to the pool — see Edge cases below). No session cookie is issued.

- **Acceptance evidence.** On Continue, server writes in a single transaction:
  - `users.tos_accepted_at` — timestamp of acceptance.
  - `users.tos_version_hash` — content hash of the ToS document the user was shown.
  - `users.privacy_version_hash` — content hash of the Privacy Policy the user was shown.
  - `users.tos_acceptance_ip` — IP address at acceptance time.
  - `users.tos_acceptance_user_agent` — User-Agent header at acceptance time.

  This is the dispute-resolution record: which versions were accepted, by which client, when. Document version hashes are computed at deployment time and frozen; rendered alongside the documents on the acceptance screen for transparency (small footer text: "ToS v1.0 · `<hash>`").
- **Mid-experiment ToS / Privacy Policy updates.** If the lawyer issues a revised ToS or Privacy Policy mid-experiment, the new version's hash differs from the user's stored hash. **No automatic re-prompt in v1** — the user remains on the version they accepted at signup. ADR-TOS-UPDATE governs the policy for any mid-experiment revision: whether existing users must re-accept, whether continued use constitutes acceptance, and how the change is communicated. In v1 we ship one ToS version and assume no mid-experiment revisions are needed (Q5 finalisation pre-launch); the ADR mechanism exists for exception cases.

- **Edge cases.**
  - *User closes the tab mid-flow.* The F-AUTH-3 `users` row remains with `tos_accepted_at IS NULL` and the `(colour, animal, number)` tuple stays assigned to that row. On the user's next sign-in attempt with the same Google account or email, F-AUTH-1 / F-AUTH-2 finds the existing `users` row, but the auth middleware sees `tos_accepted_at IS NULL` and routes the user back to F-AUTH-4 — same identity, same screen, fresh acceptance attempt. No second pool consumption.
  - *User signs up multiple times before accepting.* Each F-AUTH-1 / F-AUTH-2 attempt that finds an existing `users` row routes back to F-AUTH-4 without re-running F-AUTH-3. The pool is not consumed twice.
  - *Tab race.* User opens two tabs of F-AUTH-4 for the same `users` row. Both tabs show identical pseudonym + PFP. Whichever tab clicks Continue first writes the acceptance row. The second tab's Continue click is a no-op idempotent acceptance — server sees `tos_accepted_at IS NOT NULL` and returns the existing session cookie. No double-write.
  - *Stale unaccepted users.* `users` rows with `tos_accepted_at IS NULL` older than 30 days are purged by a daily admin sweep, releasing the linked `(colour, animal, number)` tuple back to the pool. This caps the pool drainage from abandoned signups. The 30-day window is operational tuning, not number-tuning-pass.

- **Response.** Session cookie issued. User redirected to the post-signup landing page (market list / debate view).
- **Errors.** 400 `tos_acceptance_required` (Continue pressed without checkbox ticked — should be UI-prevented but server-side check exists). 410 `tos_version_changed` (ToS document hash changed between page load and Continue press — re-renders the screen with the new version, defensive against deploy-during-signup races).
- **Acceptance.** `tests/server/auth/tos.test.ts::warning-rendered-emphasised`, `tests/server/auth/tos.test.ts::pseudonym-and-pfp-shown-as-permanent`, `tests/server/auth/tos.test.ts::tos-and-privacy-rendered-inline`, `tests/server/auth/tos.test.ts::checkbox-required-before-continue`, `tests/server/auth/tos.test.ts::acceptance-evidence-recorded`, `tests/server/auth/tos.test.ts::cancel-leaves-tos-null`, `tests/server/auth/tos.test.ts::reentry-routes-back-to-tos-without-pool-reconsumption`, `tests/server/auth/tos.test.ts::tab-race-idempotent-acceptance`, `tests/server/auth/tos.test.ts::stale-unaccepted-users-swept-after-30d`.

> ***UI/UX phase note.*** *This flow specifies the structural requirements: which elements appear, what acceptance evidence is recorded, how edge cases resolve. The visual treatment — typography, spacing, the exact dimensions of the scrollable regions, mobile-vs-desktop layout, microcopy, button colour and placement — is deferred to the UI/UX phase. The structural commitments above (inline-rendered ToS + Privacy Policy on the same screen as the checkbox; H4 warning emphasised; pseudonym + PFP rendered as permanent; single combined checkbox; acceptance evidence captured) are spec-locked and cannot be relaxed by the UI/UX pass without an ADR amending this section.*

F-AUTH-ADMIN — Admin sign-in (route-gated, structurally separate from participants)

The admin (Hrishikesh, single-admin per `E4`) authenticates via a dedicated path that is structurally outside the participant identity system. The admin has no `users` row, no pseudonym, no PFP, no ToS acceptance gate, and cannot reach any participant write surface. This is the structural enforcement of `B5`: admin is not a participant by data-model construction, not by runtime check.

- **Pre.** Admin navigates directly to `/admin/login` (URL not linked from any public surface). Discovered out-of-band; no public navigation entry point.
- **System.**
  1. `/admin/login` renders a single "Sign in with Google" button.
  2. On click, redirect to Google OAuth 2.0. On callback, server compares the authenticated email address against the env var `ADMIN_EMAIL`.
  3. **Match:** server creates a row in the `admin_sessions` table (`session_id`, `issued_at`, `last_seen_at`), sets the admin-session cookie (separate cookie name from participant session — e.g., `zugzwang_admin_session`), and redirects to `/admin` (the Control Centre hub).
  4. **No match:** server rejects with 403 `admin_email_not_allowlisted`. No row written to `admin_sessions`. No partial state. The error page does not reveal whether the email is correct or which email is allowlisted.
- **Response.** Admin session established; redirect to `/admin`.
- **Errors.** 400 `oauth_callback_error` (Google returned malformed token), 403 `admin_email_not_allowlisted`, 500 `admin_session_persistence_failed`.
- **Auth-flow boundaries.**
  - Admin sign-in does **not** create or touch a `users` row. The admin cannot accidentally end up with a participant identity.
  - Admin session does **not** grant participant powers. The participant write paths (F-BET-1, F-COMMENT-1, F-COMMENT-6, etc.) require a participant session and a `user_id` — admin has neither.
  - Participant session does **not** grant admin powers. The Control Centre routes (`/admin/*`) and the inline admin affordances on public pages check for a valid `admin_sessions` row server-side; participant-session cookies are ignored at admin endpoints.
  - The two cookie types have different names and are issued / validated independently. A user who is also the admin (i.e., the same human) can hold both cookies in the same browser session — they would log in twice, once via F-AUTH-1 / F-AUTH-2 if they wanted to participate (which `B5` forbids; this is hypothetical and surfaces a constraint), once via F-AUTH-ADMIN to operate. In practice, the admin does not hold a participant session at all.

- **Backup-admin / loss-of-access.** Recovery is an ops action, not a database action: the admin (or `BREAK_GLASS.md` recipient) updates the `ADMIN_EMAIL` env var to a new allowlisted email and redeploys. Existing admin sessions for the prior email are invalidated by the next session-validation call when the cookie's `session_id` is checked against `admin_sessions` and the actor's email no longer matches the allowlist (server-side check on every admin request).
- **Acceptance.** `tests/server/auth/admin-login.test.ts::allowlisted-email-creates-session`, `tests/server/auth/admin-login.test.ts::non-allowlisted-rejected-403`, `tests/server/auth/admin-login.test.ts::no-users-row-touched`, `tests/server/auth/admin-login.test.ts::admin-cookie-separate-from-participant`, `tests/server/auth/admin-login.test.ts::participant-session-does-not-reach-admin-routes`, `tests/server/auth/admin-login.test.ts::admin-session-does-not-reach-participant-write-paths`.

F-AUTH-5 — Logout (manual, the only session-end path)

Applies to both participant sessions and admin sessions. Each session type has its own logout endpoint and clears its own cookie; there is no cross-type logout (logging out of a participant session does not log out an admin session in the same browser, and vice versa).

- **Pre.** Authenticated user (participant or admin) clicks `Log out`.
- **System.** Single transaction: delete the server-side session row keyed by the cookie's session ID — `sessions` for participants, `admin_sessions` for admin. Clear the corresponding cookie on the response (`Set-Cookie` with `Max-Age=0` and matching `Domain` / `Path` / cookie-name attributes). Subsequent requests with the now-invalid cookie hit the auth middleware, find no matching server-side session, and are treated as anonymous (or, for admin routes, redirected to `/admin/login`). **No other code path invalidates a session** — no expiry, no idle-timeout sweeper, no admin-revoke endpoint in v1. Account ban (per `E2`) is a *user-state* change that causes the auth middleware to reject the user even with a valid session, but does not delete the session row.
- **Response.** 200 with redirect to the public landing page.
- **Errors.** None expected; logout is idempotent (logging out an already-anonymous client is a no-op).
- **Acceptance.** `tests/server/auth/logout.test.ts::server-side-row-deleted`, `tests/server/auth/logout.test.ts::cookie-cleared`, `tests/server/auth/logout.test.ts::subsequent-requests-anonymous`.

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§14 Moderation

Thesis relevance: (c) legal/safety floor.

Per `cluster_e_decisions.md` and `spec1_section_moderation.md`. **Pattern locked; thresholds, vendor selection, and edge-case behaviour finalised after Aug 15–31 sample-content testing. Final close target: 2026-09-01.**

Three tracks at submission. Two AI services run in parallel: text (OpenAI moderation API) and image (CSAM hash via PhotoDNA-or-equivalent + general adult/violence/weapons classifier, vendor TBD).

TRACK	AI FLAG CATEGORIES	ACTION	ADMIN IN LOOP?
A	CSAM, sexual minors	Block + auto-report (legal) + auto-ban user	No
A	NSFW / sexual content; adult imagery	Block + auto-ban user	No
B	Graphic violence; threats; hate speech; harassment	Hide from public; queue for admin: **Approve user** or **Block user**	Yes
C	Below threshold	Posts normally	No

Admin-in-the-loop is **Experiment-phase only**. Testnet and mainnet replace it with community-driven moderation. Moderation removes content out-of-bounds for any market discourse — *not* content wrong about a market. No misinformation track (would violate the thesis).

F-MOD-1 — Track A auto-ban

- **System.** Comment never reaches public view. User account flagged `banned` (per `E2`). Append-only `mod_actions` row written. Existing positions ride to resolution. Existing comments preserved with B1 markers frozen at ban-time. No daily allowance from ban-time forward. No appeal in v1. CSAM specifically: legal report filed automatically.
- **Acceptance.** `tests/server/moderation/track-a.test.ts::auto-ban-and-positions-preserved`.

F-MOD-2 — Track B admin review

- **System.** Comment hidden from all public surfaces. Author sees `Comment under review` on their own profile only. Admin queue at `/admin/moderation` sorted oldest-first. Each row shows comment, AI flag categories with confidence scores, user pseudonym, user Dharma, user prior-flag count.
- **Acceptance.** `tests/server/moderation/track-b.test.ts::queue-and-visibility`.

F-MOD-3 — Admin Approve / Block decision

- **System. Approve user**: comment restored to public view, user untouched, flag closed and logged. **Block user**: comment stays hidden, user banned (Track A mechanics), flag closed and logged. No appeal. No middle option (no warn-and-restore, no edit-and-resubmit) in v1. Two surfaces, same backend: the hub queue at `/admin/moderation` (two buttons per row) and the inline affordance on debate views (per F-ADMIN-4 in §15). Both paths call the same endpoint, write the same `mod_actions` row, and apply the same audit discipline.
- **Acceptance.** `tests/server/moderation/track-b.test.ts::approve-and-block-paths`.

F-MOD-4 — Bet+comment atomicity on flag (entry case)

- **Pre.** F-BET-1 entry transaction. Entry comment runs through moderation.
- **System.** Provisional decision: **atomic.** If entry comment is Track A or Track B, the entire bet+comment transaction aborts. User revises and retries. INV-1 preserved by failing both. Confirm or override during sample-content testing if revision creates abusive retry loops.
- **Acceptance.** `tests/server/bets/moderation-atomicity.test.ts::entry-flag-fails-both`.

F-MOD-5 — User banned mid-session

- **System.** On next request after ban-event, server returns 403. Existing session cookie remains technically valid; subsequent bet/comment writes return 403 `banned_user` from F-BET-7 / F-COMMENT-1 paths.
- **Acceptance.** `tests/server/auth/session.test.ts::banned-mid-session`.

Out-of-scope

No misinformation moderation. No community / user-side moderation in v1 (per `C9`). No NSFW warning-wrap (no product fit). No three-strike model (per `E2`). No appeal flow in v1.

Provisional gates

- Aug 15–31 sample-content testing (vendor + thresholds + admin queue throughput).
- PhotoDNA / equivalent CSAM service onboarded and verified.
- Lawyer review of ToS + content policy.

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§15 Admin Operations (Admin Control Centre)

Thesis relevance: (b) operationally enabling.

Per `B5`, `J2`, `J10`, `E3`, `E4`. Admin role is bounded: operational, not participatory.

The admin (Hrishikesh, single-admin per `E4`) is the operational moderator across all markets — analogous to a Reddit moderator, but with one strict divergence: per `B5`, admin **cannot bet, comment, vote, or hold positions**. There is no admin posting voice. The admin is purely operational; transparency is delivered via the public dataset at conclusion (per `E5`).

Admin actions reach the system through a single consolidated surface called the **Admin Control Centre**, accessed via two complementary patterns. The Centre is a UI consolidation of capabilities specified in F-ADMIN-1 through F-ADMIN-5 — it grants no new powers. Every action taken in the Centre lands in `admin_events` and `mod_actions` with the same append-only discipline as elsewhere.

The Control Centre is internal-facing — admin-only at every surface. No part of `/admin` or its sub-routes is reachable by anonymous or non-admin users. Inline admin affordances on public pages (e.g., the comment-removal icon on a debate view) render conditionally based on the viewer's session role and are gated server-side at the backend endpoints; an admin sees them, the Audience does not, and a request crafted to the underlying endpoint without a valid `admin_sessions` row is rejected at the auth middleware. The Centre may *display* the same data as public surfaces in admin form (e.g., a K_eff thumbnail on the hub homepage mirrors the public `/k-eff` dashboard), but those displays are admin-only views, not admin extensions to the public surfaces. Admin authentication is structurally separate from participant authentication (per F-AUTH-ADMIN in §13); the admin does not have a `users` row, cannot accidentally end up with participant powers, and cannot accidentally end up with admin powers from a participant session.

15.1 Two access patterns

The Control Centre is reachable two ways. Both call the same backend endpoints; both write the same audit-log rows. The difference is *where the admin enters the action from*.

Hub (`/admin`). The admin's workplace. Cross-market dashboard, mod queue, market list, audit log search. Used for orchestration — systematic work, market-level events, deliberation-required actions.

Inline. When an authenticated admin views public-facing pages (e.g., `/markets/<id>`), small admin-only affordances render alongside the content for *content-level remediation only*. The admin clicks a comment-removal icon next to a comment they're reading. The action calls the same endpoint as the hub mod queue. Used for opportunistic work — the admin notices something while reading and remediates without context-switching.

The split is principled.

- **Inline = content-level remediation.** Specifically: removing a Track A or Track B comment, approving a Track B pending comment. These are small, targeted, comment-scoped actions where the admin is already reading the comment in context.

- **Hub = market-level orchestration.** Everything else. Market creation, pool seeding, resolution triggering, market voiding (per F-RESOLVE-3 in §11), user banning, audit log search. These actions have larger consequences (INV-4 lock-in, full bet refunds, account-state changes) and the deliberation matters. Hub forces context.

The exact set of inline admin affordances on a market page is: remove a comment, approve a Track B pending comment. **Nothing else.** No “trigger resolution” button on the market page even when state=Closed. No “void market” button. No “ban this user” button on the comment author’s pseudonym. No bulk-action selector. Those route the admin to the hub.

Server-side enforcement. The inline affordances are rendered conditionally based on the admin-session cookie’s presence; the *backend* endpoints independently verify a valid `admin_sessions` row on every request. A client-manipulated request without admin authentication is rejected at the endpoint, not at the UI layer. Inline is a UI affordance, not a privilege escalation path.

15.2 The hub homepage

The `/admin` route lands on a dashboard surfacing what needs the admin’s attention. Five widgets:

1. **Mod queue.** Count of Track B comments currently held under review, with age-of-oldest. Single chronological queue across all markets — global view only, no per-market filter.
2. **Signup rate + identity-pool depth.** Recent 24h / 7d signup counts. Current `identity_pool` unassigned count as a percentage of total. Alarm threshold visible: red when below 5%.
3. **K_eff thumbnail.** Live K_eff trajectory chart, small format. Click-through to the full public dashboard at `/k-eff`. The thumbnail is admin-only (it lives inside the Control Centre); the full dashboard at `/k-eff` is the public surface (per `J5`, `G3`) — same data, two surfaces.
4. **Market state counts.** Count of markets in each state: `Draft / Open / Closed / Resolving / Resolved / Voided`. Click-through to the market list.
5. **Suspicious-pattern indicators.** Per-user bet velocity outliers, comment velocity outliers, multiple-account-same-IP flags. Heuristic only — no automated bans (per F-ADMIN-5). Admin reviews and decides via the user record in the moderation tab.

Widgets render server-side at page load; no live websocket. Refresh on navigation. Per `C7`, polled-on-view is the platform’s general default; the admin hub follows the same rule.

15.3 Tab structure

Three tabs, deliberately minimal:

- **Home** (`/admin`) — the homepage above. The default landing.
- **Markets** (`/admin/markets`) — market list with state filters; entry point to F-ADMIN-1, F-ADMIN-2, F-ADMIN-3.

- **Moderation** (`/admin/moderation`) — global mod queue (F-ADMIN-4) and audit log search (F-ADMIN-5).

Analytics is folded into the Home dashboard. There is no separate Analytics tab. If the admin needs deeper analytics than the homepage shows, they query the database directly outside the product surface.

F-ADMIN-1 — Market creation (Hub: Markets tab)

- **Pre.** Admin authenticated (valid `admin_sessions` row), on `/admin/markets/new`.
- **System.** Admin enters question, resolution criterion, deadline. **Form validates `deadline ≤ 2026-11-05 23:59 UTC`** (per `J10`, `B8`). Curation policy is admin discretion in Experiment phase (per `J2`). Optional `display_order` field for hand-curated market list (per `C4`).
- **Response.** Market in `Draft` state.
- **Surface.** Hub only. No inline equivalent.
- **Acceptance.** `tests/server/admin/markets.test.ts::deadline-form-validation`.

F-ADMIN-2 — Pool seed (Hub: Markets tab)

- **Pre.** Market in `Draft`. Admin commits to open it.
- **System.** Admin enters seed magnitude. Single transaction: `pool_seed` flow from the admin account (a synthetic actor in the Dharma ledger; not a `users` row) → pool reserves. Initial price determined by reserve split (50/50 by default for binary markets). Market transitions `Draft → Open`. Per `B5`, criterion-based sizing — number-tuning pass owns specific magnitude.
- **Response.** Market `Open` with seeded pool.
- **Surface.** Hub only. No inline equivalent.
- **Acceptance.** `tests/server/admin/pool-seed.test.ts::seed-flow-and-state-transition`.

F-ADMIN-3 — Resolution trigger (Hub: Markets tab)

- **Pre.** `market.state = Closed`. Admin determines outcome.
- **System.** Admin selects winning side, attaches resolution evidence (URL or text). Market transitions `Closed → Resolving`. F-RESOLVE-1 then runs.
- **Response.** Resolution event ID.
- **Surface.** Hub only. Resolution is a market-level event with permanent INV-4 consequences; no inline button on the market detail page even when state=Closed. Admin must navigate to the hub to act.
- **Acceptance.** `tests/server/admin/resolution.test.ts::resolving-state-then-resolved`, `tests/server/admin/resolution.test.ts::no-inline-resolution-affordance`.

F-ADMIN-4 — Moderation actions (Hub: Moderation tab; Inline: market pages)

- **Pre.** Track B comment exists with `review_status = pending`.
- **System.** Admin reviews comment context (text, image, AI flag categories with confidence, user pseudonym, user Dharma, user prior-flag count). Two actions available:
  - **Approve user** — comment restored to public view, user untouched, flag closed and logged in `mod_actions`.
  - **Block user** — comment stays hidden, user banned (Track A mechanics per `E2`), flag closed and logged.
- **Response.** Comment state updated; `mod_actions` row written; if Block, `users.banned_at` set.
- **Surface — hub.** `/admin/moderation` shows the global queue, sorted oldest-first. Approve / Block buttons per row. Used for systematic queue work.
- **Surface — inline.** When an authenticated admin views `/markets/<id>` or any debate view, Track B pending comments render inline (visible to admin only, with a yellow `pending review` badge) alongside Approve / Remove buttons. Approving from inline restores the comment to public view; Remove keeps it hidden and bans the author. Same backend endpoints as the hub. **Server independently verifies a valid `admin_sessions` row on every request — the inline affordance is a UI convenience, not a privilege path.**
- **Inline scope explicitly:** approve a Track B pending comment, remove a Track A or Track B comment. Nothing else. No "ban user" button on the comment author's pseudonym (banning is hub-only — accessed via the moderation queue or a user-record action). No "trigger resolution" button on the market header. No bulk-select.
- **Acceptance.** `tests/server/admin/moderation.test.ts::hub-queue-approve-block`, `tests/server/admin/moderation.test.ts::inline-remove-pending-comment`, `tests/server/admin/moderation.test.ts::inline-affordance-admin-only-server-verified`, `tests/server/admin/moderation.test.ts::inline-scope-comment-only-no-user-ban-no-resolve`.

F-ADMIN-5 — Audit log search (Hub: Moderation tab)

- **Pre.** Admin on `/admin/moderation` (audit-search sub-surface).
- **System.** Per `E3`. Search across `admin_events` and `mod_actions` rows by date range, action type, market, user, or pseudonym. Heuristic; no automated bans on velocity-flag results.
- **Response.** Filtered list of events.
- **Surface.** Hub only. Audit log search is investigation work; no inline analogue makes sense.
- **Acceptance.** `tests/server/admin/audit-search.test.ts::query-by-date-action-market-user`.

Single-admin assumption (per `E4`)

No backup admin in v1. Mitigations: append-only audit trails (no real-time presence required), buffered resolution windows, mod queue tolerant to 12–24h latency. `BREAK_GLASS.md` runbook is a Hrishikesh-owned, non-code action item — sealed-envelope credentials handoff for a trusted second person, plus the operational recovery path: update `ADMIN_EMAIL` env var and redeploy (per F-AUTH-ADMIN backup-admin clause).

What the Control Centre is NOT

Explicit out-of-scope for the Centre, beyond the standard §18 list:

- **No emergency override of INV-4.** Resolved markets cannot be re-opened, edited, or un-resolved through the Centre. Corrections happen via F-RESOLVE-2 (compensating events) only.
- **No silent shadow-ban or soft-suppress.** Every moderation action writes a `mod_actions` row. There is no UI mechanism to hide a user's content from public view without an audit row.
- **No bulk-action API or multi-select.** Mod actions are per-comment. If the admin wants to remove ten comments, they action ten removals.
- **No participatory powers.** Admin cannot bet, comment, vote, hold positions, or post under an admin voice. Per `B5`, enforced structurally — admin has no `users.id` (per F-AUTH-ADMIN).
- **No public mod log surface.** `admin_events` and `mod_actions` are admin-only audit tables during the experiment; full release at conclusion per `E5`. No live or hub-tab log surface.
- **No multi-admin role layers.** Single admin in v1; multi-admin readiness deferred to testnet phase.

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§16 Operational Floor

Thesis relevance: (b) operationally enabling, (c) legal/safety floor.

Consolidates Constants, Errors, Privacy, Audit Logs, and Compliance as numbered subsections. Each is brief by design — these are world-standard treatments, not sites of innovation.

16.1 Constants and Limits

Symbolic only. Specific values → number-tuning pass (target: 2026-09-01).

CONSTANT	WHY
`INITIAL_USER_DHARMA`	Signup balance for every user account.
`ADMIN_INITIAL_DHARMA`	Operational seed for admin account, abundantly sized.
`DAILY_ALLOWANCE_DHARMA`	Daily fresh-allowance accrual per UTC day.
`POOL_SEED_PER_MARKET`	Default pool seed; criterion: typical trade → small price impact, cumulative activity → meaningful drift.
`BET_MIN_STAKE`	Minimum stake per bet.
`COMMENT_MAX_LENGTH`	Maximum comment text length.
`REPLY_DEPTH_MAX`	Maximum reply nesting depth.
`SLIPPAGE_WARNING_PCT_THRESHOLD`	Threshold above which slippage modal triggers (per `G1`).
`IN_FLIGHT_BET_TIMEOUT_SEC`	Window during which `Resolving`-state in-flight bets may commit (per `G6`).
`POLL_INTERVAL_MS_DEBATE_VIEW`	Debate-view poll interval (per `C7`).
`OTP_TTL_MIN`	Email-OTP time-to-live (per `K1`).
`OTP_REQUESTS_PER_EMAIL_PER_HOUR`	Per-email OTP request cap (anti-spam / anti-bot).
`OTP_REQUESTS_PER_IP_BURST_PER_MIN`	Per-IP OTP request burst cap.
`RATE_LIMIT_PER_MARKET_PER_DAY`	Per-user, per-market write cap covering direct comments, replies, image-comments, and friendly-fire votes (shared budget per §8). Bets are not bound by this cap.

CONSTANT	WHY
`RATE_LIMIT_BURST_PER_MIN`	Per-user burst cap covering the same shared comment / friendly-fire write surface.
`AI_FLAG_THRESHOLD_TRACK_A_*`	Per-category Track A confidence thresholds (`E1`).
`AI_FLAG_THRESHOLD_TRACK_B_*`	Per-category Track B confidence thresholds (`E1`).

16.2 Error Handling and Failure Modes

Each F-flow already names its errors inline. This subsection is a cross-reference for shared codes.

TRIGGER	USER SEES	SYSTEM DOES
Insufficient Dharma (pre-submit)	Form-level error with current balance + required stake.	No transaction opened.
Market closed mid-bet	"Market closed at HH:MM:SS UTC".	400 `market_closed_at`; transaction aborted.
Comment auto-banned (Track A)	Entry bet rejected; comment-revision UI prompts retry without indicating *which* category fired.	F-MOD-4 atomicity; mod-action row written; user banned.
Comment held in admin review (Track B)	"Comment under review" surface on user's *own* profile only.	Hidden from public; queue row written.
User banned mid-session	403 with notice; session cookie remains but write paths reject.	F-MOD-5.
AI moderation false-positive ban	No appeal in v1. User contacts admin via `foundation@zugzwangworld.com`. Admin can manually reverse via append-only correction event.	Mitigation owned by sample-content testing + threshold tuning.
Slippage above threshold	Confirmation modal with from/to prices.	Per `G1`; transaction proceeds on confirm.
Network failure on bet	Idempotency key prevents duplicate submission.	Client retries on key-aware endpoint.
Erasure-request (banned or active user)	"30-day verification window pending".	Per `H2`; pseudonym scrubbed at window end.
Admin queue overload	Latency on Track B reviews increases.	Mitigation: tighten thresholds, narrow categories — sample-content testing window flags this.

TRIGGER	USER SEES	SYSTEM DOES
Pool solvency	Structurally guaranteed by CPMM math.	Not a runtime failure mode.

16.3 Privacy and Data

Operating principle (per `Cluster_B`): **transparency-by-design.** Public visibility is the default; the legal floor reconciles via pseudonym-scrub.

- **Public-by-default surface (`D7`).** Full debate view, pseudonyms, bets, stakes, Dharma values, comments, leaderboard, K_eff dashboard — all visible to unauthenticated users. Bet placement, comment posting, profile editing, erasure requests are gated to authenticated users.
- **Profile pages (`D8`).** Pseudonym, complete bet history, complete comment history with markers preserved, current Dharma balance, daily-allowance usage history (private to user), join date, optional self-declared X handle, win/loss aggregate stats. Banned users: profile shows `Banned` label visible to all; bet/comment history remains visible.
- **Right-to-erasure (`H2`).** Pseudonym-scrub model: on valid request, pseudonym replaced with permanent placeholder (e.g., `[removed_user_4721]`); PII fields wiped (email, Google ID, IP, user-agent, self-declared X handle); bets, comments, mod-actions, ledger rows persist under placeholder. 30-day verification window. Banned users may request scrub.
- **Re-identification warning (`H4`).** Re-id risk is transparency-by-design, not a bug to mitigate. ToS warning at signup quoted in F-AUTH-4.
- **Session-level logging (`H3`).** Structured request log per request: timestamp, user_id (or anon marker), route, status_code, IP, user_agent, latency_ms. **No request body. No response body.**
- **Public dataset (`H1`, `E5`).** Released 2026-11-06 — see §12.2.

Lawyer-review batch (Hrishikesh-owned action item, pre-launch): ToS wording for transparency-by-design, pseudonym-scrub, re-id warning; erasure-request flow + jurisdictional SLAs (DPDPA 30d / GDPR 1mo / CCPA 45d); cross-border data residency (R2 buckets, Supabase regions); content policy section (E1 tracks); grievance officer / compliance contact (DPDPA basics).

16.4 Audit Logs

Append-only ledgers underpinning the public dataset. Enforcement is at row-level rule + audit trigger; `UPDATE` is rejected.

TABLE	ROWS	VISIBILITY (DURING EXPERIMENT)	PUBLIC RELEASE
`dharma_ledger`	Every Dharma flow with tag from fixed enum.	Per-user views on profile.	Full at 2026-11-06.

TABLE	ROWS	VISIBILITY (DURING EXPERIMENT)	PUBLIC RELEASE
`bets`	Every bet ever placed.	Public via debate view + profile.	Full at 2026-11-06.
`comments`	Every comment ever posted (excluding Track A removed).	Public via debate view + profile.	Full at 2026-11-06 (Track A removed media not released).
`resolution_events`	Resolution + correction + void events.	Public on market detail page.	Full at 2026-11-06.
`payout_events`	Per-bet settlement entries.	Public via profile.	Full at 2026-11-06.
`mod_actions`	Every Track A and Track B action with reason, AI category, confidence, override history.	Admin-only during experiment (per `E5`).	Full at 2026-11-06.
`admin_events`	Every admin-initiated state change (market creation, resolution trigger, void, mod action). `actor_id` references the admin allowlist identity (constant `'admin-singleton'` in v1 per E4) — not `users.id`. The admin is structurally outside the `users` table per F-AUTH-ADMIN.	Admin-only during experiment (per `E5`).	Full at 2026-11-06.
`user_events`	Signup, ban, scrub, daily-allowance accrual.	Per-user views on profile.	Full at 2026-11-06.
`friendly_fire_events`	Every friendly-fire vote with `voter_id, comment_id, market_id, comment_side, direction, voter_side_at_cast_time, timestamp, frozen_at` (NULL until exit-to-zero, then set; never deleted). Live display counts exclude frozen votes (per §8 F-COMMENT-8); audit log retains all.	Public counts visible per §9; full row not visible during experiment.	Full at 2026-11-06.

TABLE	ROWS	VISIBILITY (DURING EXPERIMENT)	PUBLIC RELEASE
`sessions`	Server-side participant session rows backing the indefinite session cookie. Row written on F-AUTH-1 / F-AUTH-2 success. Row deleted on F-AUTH-5 (manual logout); no other deletion path. Stores: `session_id, user_id, issued_at, last_seen_at`. Not append-only — `last_seen_at` updates on every authenticated request.	Not visible to users or to the Audience. Admin-only diagnostic view via direct DB query.	Not released — sessions table excluded from public dataset (no thesis-relevant signal; user-correlatable session activity is privacy-sensitive).
`admin_sessions`	Server-side admin session rows backing the admin-session cookie. Row written on F-AUTH-ADMIN success. Row deleted on F-AUTH-5 admin logout; no other deletion path. Stores: `session_id, admin_email, issued_at, last_seen_at`. Single row at any moment in v1 (single admin per E4). Validation on every admin request also re-checks the actor's email against `ADMIN_EMAIL` env var, so an env-var change implicitly invalidates active sessions for non-allowlisted emails.	Not visible to users, to the Audience, or to the admin via product surface. Admin-only diagnostic view via direct DB query.	Not released — admin session activity is operational, not thesis-relevant.
`identity_pool`	Pre-generated `(colour, animal, number, pfp_filename, assigned_at)` tuples backing F-AUTH-3 pseudonym + PFP assignment. Inserted in bulk pre-launch via the asset pipeline (see §13 F-AUTH-3). `assigned_at` updated from NULL to a timestamp when a tuple is consumed by signup. Not append-only — `assigned_at` is the single permitted update per row.	Not visible to users. Admin-only operational view (pool depth, exhaustion alerting).	Not released — operational table; the assignment of pseudonym to user is implicitly visible via the `users` row, which IS released.

Append-only enforcement on `mod_actions`, `admin_events`, and `friendly_fire_events` parallels INV-4 mechanics — these are not §5 invariants, but the same row-level rules and tests apply. The single permitted update on `friendly_fire_events` is setting `frozen_at` from NULL to a timestamp on exit-to-zero (per F-COMMENT-8); content fields are immutable.

16.5 Compliance

- **AGPL-3.0 §13 source link (`I6`).** Footer text on every page: `Source code available at github.com/zugzwang-foundation/experiment under AGPL-3.0`. Hard legal requirement. Five-minute task. Owner: Hrishikesh, before launch.
- **Terms of Service.** Drafted; lawyer-finalised wording. Includes transparency-by-design statement, pseudonym-scrub model, re-identification warning, content policy referencing §14, ToS-implicit consent for K2 brand-account quoting.
- **Privacy Policy.** Drafted; lawyer-finalised. Cross-references §16.3.
- **DPDPA grievance contact.** Email + named officer (per Indian Digital Personal Data Protection Act). Surfaced in footer and Privacy Policy.
- **PhotoDNA + reporting compliance.** CSAM hash service onboarded pre-launch. Auto-report to NCMEC (or jurisdictional equivalent) on Track A CSAM detection.
- **DSA notice-and-action.** Small-platform exposure noted. ToS includes notice-and-action contact + procedure.

Lawyer engagement target: mid-July 2026. ToS / Privacy Policy drafts ready four weeks before launch.

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§17 Acceptance Tests

Thesis relevance: (b) operationally enabling.

Flat list. Each entry: **test name → spec section it covers → invariants enforced.** Every flow above and every invariant must have at least one test here; orphans are suspect.

TEST	SECTION	INVARIANTS
`bet-comment-atomicity::happy-path-entry`	§7 F-BET-1	INV-1, INV-3
`bet-comment-atomicity::rolls-back-on-comment-fail`	§5 INV-1	INV-1
`bet-comment-atomicity::rolls-back-on-bet-fail`	§5 INV-1	INV-1
`bet-comment-atomicity::rejects-missing-fields-400`	§5 INV-1	INV-1
`dharma-non-transferable::no-user-to-user-write`	§5 INV-2	INV-2
`dharma-non-transferable::ledger-tag-required`	§5 INV-2	INV-2
`side-frozen::after-author-flips`	§5 INV-3	INV-3
`side-frozen::after-author-exits`	§5 INV-3	INV-3
`side-frozen::reply-inherits-replier-side`	§5 INV-3	INV-3
`side-frozen::no-position-no-new-comment`	§8 F-COMMENT-5	INV-3
`resolution-append-only::rejects-update`	§5 INV-4	INV-4
`resolution-append-only::correction-new-event`	§11 F-RESOLVE-2	INV-4
`resolution-append-only::clawback-floors-zero`	§11 F-RESOLVE-2, §10.7	INV-4, INV-2

TEST	SECTION	INVARIANTS
`resolution-append-only::correction-no-comment-unlock`	§5 INV-4	INV-4
`subsequent-buy::happy-path`	§7 F-BET-2	INV-2
`sell::preserves-comments`	§7 F-BET-3	INV-3
`bet-validation::insufficient-dharma`	§7 F-BET-4	—
`bet-race::closed-mid-bet`	§7 F-BET-5	—
`bet-race::in-flight-resolving`	§7 F-BET-6	INV-1
`bet-auth::banned-user-rejected`	§7 F-BET-7	—
`bet-slippage::warning-modal`	§7 F-BET-9	—
`bet-single-side::opposite-side-rejected`	§7 F-BET-10	—
`comment-direct::position-required`	§8 F-COMMENT-1	INV-3
`comment-reply::replier-side`	§8 F-COMMENT-2	INV-3
`comment-media::moderation-routes`	§8 F-COMMENT-3	INV-1, INV-3
`comment-validation::length-limit`	§8 F-COMMENT-4	—
`debate-view::ranking-function-order`	§9 F-DEBATE-1	—
`debate-view::replies-chronological-ascending`	§9 F-DEBATE-1	—
`debate-view::friendly-fire-live-count`	§9 (preamble), §8 F-COMMENT-8	—
`friendly-fire::cross-aisle-up-same-side-down`	§8 F-COMMENT-6	—
`friendly-fire::self-vote-rejected`	§8 F-COMMENT-6	—

TEST	SECTION	INVARIANTS
`friendly-fire::already-voted-rejected`	§8 F-COMMENT-6	—
`friendly-fire::clear-recomputes-display`	§8 F-COMMENT-7	—
`friendly-fire::freeze-on-exit`	§8 F-COMMENT-8	—
`rate-limit::shared-budget-comments-and-friendly-fire`	§8 (preamble), §16.1	—
`debate-view::three-state-marker`	§9 F-DEBATE-2	INV-3
`debate-view::marker-frozen-at-resolution`	§9 F-DEBATE-3	INV-3, INV-4
`debate-view::poll-interval`	§9 F-DEBATE-4	—
`resolution::settles-and-locks`	§11 F-RESOLVE-1	INV-2, INV-4
`resolution::void-full-refund-pool-unwind`	§11 F-RESOLVE-3	INV-2, INV-4
`auth::google-no-captcha`	§13 F-AUTH-1	—
`auth::google-existing-user-match`	§13 F-AUTH-1	—
`auth::google-new-user-routes-to-pseudonym`	§13 F-AUTH-1	—
`auth::otp-turnstile-required`	§13 F-AUTH-2	—
`auth::otp-ttl-respected`	§13 F-AUTH-2	—
`auth::otp-rate-limited`	§13 F-AUTH-2	—
`auth::pseudonym-auto-assigned-permanent`	§13 F-AUTH-3	—
`auth::pseudonym-pfp-coherent-with-name`	§13 F-AUTH-3	—
`auth::pseudonym-pool-fifo-selection`	§13 F-AUTH-3	—

TEST	SECTION	INVARIANTS
`auth::pseudonym-pool-no-double-assignment-under-concurrency`	§13 F-AUTH-3	—
`auth::pseudonym-pool-exhaustion-503`	§13 F-AUTH-3	—
`auth::tos-warning-rendered-emphasised`	§13 F-AUTH-4	—
`auth::tos-pseudonym-and-pfp-shown-as-permanent`	§13 F-AUTH-4	—
`auth::tos-and-privacy-rendered-inline`	§13 F-AUTH-4	—
`auth::tos-checkbox-required-before-continue`	§13 F-AUTH-4	—
`auth::tos-acceptance-evidence-recorded`	§13 F-AUTH-4	—
`auth::tos-cancel-leaves-tos-null`	§13 F-AUTH-4	—
`auth::tos-reentry-routes-back-without-pool-reconsumption`	§13 F-AUTH-4	—
`auth::tos-tab-race-idempotent-acceptance`	§13 F-AUTH-4	—
`auth::tos-stale-unaccepted-users-swept-after-30d`	§13 F-AUTH-4	—
`auth::logout-server-side-row-deleted`	§13 F-AUTH-5	—
`auth::logout-cookie-cleared`	§13 F-AUTH-5	—
`auth::logout-subsequent-requests-anonymous`	§13 F-AUTH-5	—
`auth::session-indefinite-no-expiry`	§13 (preamble)	—
`auth::admin-allowlisted-email-creates-session`	§13 F-AUTH-ADMIN	—

TEST	SECTION	INVARIANTS
`auth::admin-non-allowlisted-rejected-403`	§13 F-AUTH-ADMIN	—
`auth::admin-no-users-row-touched`	§13 F-AUTH-ADMIN	—
`auth::admin-cookie-separate-from-participant`	§13 F-AUTH-ADMIN	—
`auth::participant-session-does-not-reach-admin-routes`	§13 F-AUTH-ADMIN	—
`auth::admin-session-does-not-reach-participant-write-paths`	§13 F-AUTH-ADMIN	—
`moderation::track-a-auto-ban`	§14 F-MOD-1	—
`moderation::track-b-queue`	§14 F-MOD-2	—
`moderation::track-b-approve-block`	§14 F-MOD-3	—
`moderation::entry-flag-fails-both`	§14 F-MOD-4	INV-1
`moderation::banned-mid-session`	§14 F-MOD-5	—
`admin::market-deadline-form-validation`	§15 F-ADMIN-1	—
`admin::pool-seed-flow`	§15 F-ADMIN-2	INV-2
`admin::resolution-trigger`	§15 F-ADMIN-3	INV-4
`admin::no-inline-resolution-affordance`	§15 F-ADMIN-3	INV-4
`admin::moderation-hub-queue-approve-block`	§15 F-ADMIN-4	—
`admin::moderation-inline-remove-pending-comment`	§15 F-ADMIN-4	—
`admin::moderation-inline-affordance-admin-only-server-verified`	§15 F-ADMIN-4	—

TEST	SECTION	INVARIANTS
`admin::moderation-inline-scope-comment-only-no-user-ban-no-resolve`	§15 F-ADMIN-4	—
`admin::audit-search-query-by-date-action-market-user`	§15 F-ADMIN-5	—
`admin::control-centre-internal-only-anon-rejected`	§15 (preamble)	—
`admin::control-centre-internal-only-non-admin-user-rejected`	§15 (preamble)	—
`dataset::zugzwang-condition-derivable`	§3 G1, §12.2	—
`dataset::archive-bundle-released-on-time`	§3 G2, §12.2	—
`dashboard::k-eff-live-throughout`	§3 G3, §10.8	—
`pseudonym-scrub::preserves-ledger`	§3 G6, §16.3	—
`admin-not-participant::all-paths-reject`	§3 G7, §15	—
`dataset-propagation-derivable`	§3 G8, §16.4	—
`freeze::nov-5-23-59-utc-no-writes`	§12.1	—

A `pnpm test:invariants` script runs the INV-tagged subset in isolation. CI fails if any INV test fails or is skipped.

§18 Out of Scope (Negative Space)

Thesis relevance: (b) operationally enabling.

Explicit. Every item has a one-line "why not now" so Claude does not re-litigate.

- **Testnet / mainnet phases.** Different repos, different playbooks. Do not invent blockchain primitives in this codebase (per `CLAUDE.md` §1).
- **Native mobile apps.** Web responsive only.
- **End-to-end encryption.** Hosted experiment; transparency-by-design defeats E2E framing.
- **Federation.** Single deployment, single domain.
- **Per-user posting integrations to external platforms.** Brand-account propagation is admin-curated only (per `K2`).
- **Notifications (email or in-app).** Deferred to a separate workstream (per `I1`). Users discover resolutions and replies by visiting pages. Risk acknowledged; retention impact accepted.
- **About / FAQ / help page.** Deferred (per `I5`); whitepaper link in footer is the v1 explainer surface. Brand looks bare; tradeoff against ship-speed accepted.
- **Search.** Markets accessed via hand-curated list and brand-account links (per `I2`).
- **Following users / markets.** Cascade into notifications (per `I3`); not v1.
- **User-side moderation (block / hide / report).** No user-side report button in v1 (per `C9`); community moderation moves to testnet+.
- **Refunds beyond market void.** No bet cancellations, no refunds, no deadline-extension refund path (per `F4`, `B8`).
- **Deadline extensions.** Forbidden (per `B8`); markets that miss their event void per `B3`.
- **Mid-market liquidity adjustments.** Pool seed fixed at creation (per `B6`).
- **SSE / WebSockets / live realtime.** Polled-on-view in v1 (per `C7`); SSE deferred to SPEC.2.
- **Community moderation.** Testnet+ scope (per `E1`).
- **Misinformation moderation.** Markets are how Zugzwang resolves wrongness; removing content because it is wrong about a market violates the thesis (per `E1`).
- **NSFW warning-wrap.** No product fit on Zugzwang; NSFW auto-bans instead (per `E1`).
- **Three-strike ban model.** Adds friction without protective value at experiment scale (per `E2`).
- **Edit-and-resubmit / warn-and-restore.** Deferred pending sample-content findings (per `E1`).
- **Body logging on requests / responses.** PII landmine, storage-cost multiplier (per `H3`).
- **i18n.** UI English only; timestamps UTC stored, browser-local rendered (per `J6`).
- **Per-user posting to X / LinkedIn / Telegram.** No per-user external posting (per `K2`); brand-account only.
- **AI smart replies / AI rewriting of user content.** Out of scope (per `K2`).
- **Anti-spam ML beyond rate limits.** Not v1.

- **Session expiry / idle timeout / sliding window.** Sessions are indefinite per §13. Logout is the only session-end path. Trade-off documented; dataset-cleanliness noise floor accepted.
- **Fresh-OTP gates on sensitive actions** (scrub-request, ban-appeal, etc.). Indefinite session is sufficient auth for any action in v1. Trade-off: a hijacked session can trigger irreversible scrub.
- **Admin session-revoke endpoint.** No tooling to invalidate a user's session externally in v1. User-side ban (per `E2`) reads at auth middleware time but does not touch the sessions table.
- **Multi-device session list / device management UI.** Users cannot see or revoke their own sessions across browsers in v1. Logging out on one browser does not log out other browsers.
- **Forgotten-password / account-recovery flows.** No passwords exist (Google OAuth or email-OTP only). If a user loses access to their email and Google account simultaneously, recovery is via admin email contact (`foundation@zugzwangworld.com`); no automated recovery in v1.
- **Admin role flag on participant accounts.** Admin is structurally separate per F-AUTH-ADMIN — no `users.role` column, no participant account with elevated permissions. The choice is deliberate (B5 enforced by data-model construction, not runtime check); reverting to a role-flag pattern would weaken B5's structural enforcement and is rejected.
- **Public mod log surface during the experiment.** No live or hub-tab log surface. `admin_events` and `mod_actions` are admin-only audit tables during the experiment; full release at conclusion per `E5`.
- **Personalised ranking.** The ranking function is universal — same inputs, same output, same rendered order for every reader at every moment (§9). No per-reader reordering, no recommender-style personalisation, no A/B variants in production.
- **Engagement-metric ranking inputs.** View counts, dwell time, scroll depth, click-through, "users like you also voted X." Not v1 inputs to the ranking function. If introduced later, must be added via ADR + made auditable from the public dataset (§9 transparency requirement).
- **Online-learned ranking weights.** Weights are hand-set in `RANKING.md`, locked at launch via ADR-RANKING (§19 Q13), changed only via ADR. No silent online tuning, no ML-learned weights.
- **User-toggleable sort variants.** Not v1. Single ranking function, single rendered order. Toggleable views (e.g., "newest first") deferred to a UI/UX ADR.
- **Filters on the debate view.** Not v1. The implicit YES / NO column split is the only built-in filter.
- **In-product analytics export tooling (chart PNG / CSV / slide-deck export).** No admin "highlight tool", no public dashboard download buttons, no admin-curated chart export. The K_eff dashboard renders live; if admin needs static analytics for external presentation, generation happens out-of-band against the public dataset. Decision: keep the product surface minimal; internal analytics are not a product feature.
- **Multi-admin readiness / backup admin.** Single admin in v1 (per `E4`); `BREAK_GLASS.md` runbook is the mitigation.
- **Voice notes.** Comment surface is text + image (per `K3`).
- **Market editing post-creation.** Markets are immutable once `Open` (per `B7`, `B8`).
- **Pinned messages / featured comments.** Not v1.
- **Synthetic Dharma / separate liquidity ledger.** Path A only (per `F5`/`F6`, `B5`).
- **Brier overlay / time-weighted bonus on Dharma award.** CPMM-native only (per `A2`).

- **Forced position liquidation on user ban.** Positions ride to resolution (per `E2`).
- **Public mod actions in real time.** Admin-only during experiment; full release at conclusion (per `E5`).
- **Open-submission market curation.** Hrishikesh-curates in Experiment phase (per `J2`); revisits at testnet.

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§19 Open Questions and Deferred Decisions

Thesis relevance: (b) operationally enabling.

Numbered. Each: question, default behaviour shipped now, decider, resolve-by date.

#	QUESTION	DEFAULT	DECIDER	RESOLVE BY
Q1	AI flag thresholds (per category, per track)	Conservative starting values; sample-content tested.	Hrishikesh + sample-content test	2026-09-01
Q2	Image classifier vendor (Rekognition / Sightengine / Hive)	TBD; selected via implementation pass.	Engineering implementation pass	Before launch
Q3	PhotoDNA / equivalent CSAM hash service onboarded	In progress; weeks-long onboarding initiated.	Hrishikesh	Before launch
Q4	Specific rate-limit values, slippage threshold, allowance, seed magnitude, comment length, OTP TTL, poll interval, in-flight timeout	All deferred; symbolic constants only in §16.1.	Number-tuning pass	2026-09-01
Q5	ToS final wording	Drafted; lawyer-review pending.	Lawyer	Mid-July 2026
Q6	First market content	TBD; refusal rule — Claude does not invent market content.	Hrishikesh alone (per `J1`)	Before launch
Q7	Notifications design	No v1 surface; users discover by visiting pages.	Separate notifications workstream	Mid-experiment
Q8	About / FAQ surface	No v1; whitepaper link in footer.	Hrishikesh, post-launch	Mid-experiment

#	QUESTION	DEFAULT	DECIDER	RESOLVE BY
Q9	Auth library implementation (NextAuth / Auth.js, Clerk, or hand-rolled)	DECIDE; ADR-0004 gating. Stack must support: Google OAuth, email-OTP with Cloudflare Turnstile siteverify integration, server-side session table, indefinite cookies, manual logout-only session-end.	Engineering	Before launch
Q10	Bet+comment atomicity behaviour on Track B flag (entry case)	Provisional: atomic — both fail.	Sample-content testing	2026-09-01
Q11	Three-state marker icons (Flipped / Exited visual)	TBD; design pass.	UI/UX design pass	Pre-launch
Q12	Cross-platform brand-account specifics (platform list, post copy, curation algorithm, AI-misfire response)	Deferred to Social workstream.	Social workstream	Pre-launch or mid-experiment
Q13	Comment ranking function (`RANKING.md`) — input set, weights, decay shape, friendly-fire term	Open-source (AGPL-3.0); locked via ADR-RANKING; deterministic; universal; auditable from public dataset inputs	Hrishikesh + engineering	Before launch
Q14	Friendly-fire fine details — cool-down post side-flip; clear-vote vs replace-vote semantics; rate-limit weight relative to comments	No cool-down in v1; one vote per (user, comment), changeable via clear-then-recast; equal weight in shared rate-limit budget	Number-tuning pass + post-launch ADR if needed	2026-09-01

#	QUESTION	DEFAULT	DECIDER	RESOLVE BY
Q15	Pseudonym + PFP asset pipeline (ADR-PSEUDONYM): word lists, Flux prompt template, sampler / seed strategy, number-compositing pipeline, R2 upload procedure	Pattern locked in §13 F-AUTH-3 (Colour + Animal + Number, 50K namespace). Specific word lists, prompt template, and pipeline scripts ship as ADR-PSEUDONYM artefact + repo code before launch.	Hrishikesh + engineering	Before launch
Q16	OTP rate-limit specific values — `OTP_REQUESTS_PER_EMAIL_PER_HOUR`, `OTP_REQUESTS_PER_IP_BURST_PER_MIN`, `OTP_TTL_MIN`	Symbolic only in spec; specific values via number-tuning pass	Number-tuning pass	2026-09-01
Q17	Mid-experiment ToS / Privacy Policy revision policy (ADR-TOS-UPDATE) — whether existing users must re-accept on version change, what counts as continued-use acceptance, communication mechanism	No automatic re-prompt in v1; assumes one ToS version ships at launch and holds. ADR mechanism exists for exception cases.	Hrishikesh + lawyer	If/when needed
Q18	Admin auth implementation specifics (ADR-ADMIN-AUTH): exact NextAuth / Auth.js (or alternative) configuration for the dedicated `/admin/login` Google OAuth path with `ADMIN_EMAIL` allowlist; `admin_sessions` table schema; admin session middleware shape; recommendation on Google Workspace hardware-key requirement for the admin email	Pattern locked in §13 F-AUTH-ADMIN (separate auth path, separate cookie, no `users` row, server-side allowlist); implementation specifics ship as ADR-ADMIN-AUTH artefact + repo code before launch.	Engineering	Before launch

Claude does not try to resolve these; it implements the default and flags the question in PR descriptions.

§20 Change Log

DATE	VERSION	SECTION	CHANGE	RATIONALE	ADR
2026-05-01	1.0.0-draft	All	Initial draft. Section structure ratified at start of SPEC.1 chat: 20 numbered sections + 2 appendices, derived from Stitch DM template (16 sections) + Zugzwang-specific additions (Dharma Economy, Conclusion Event, Operational Floor consolidating Privacy / Audit / Compliance / Operations / Constants / Errors). Four §5 invariants mirror `CLAUDE.md` §2.1–§2.4 verbatim.	Single-source-of-truth lock; gap-resolution session of 2026-04-30 absorbed.	—

DATE	VERSION	SECTION	CHANGE	RATIONALE	ADR
2026-05-03	1.0.0-draft	§1, §2, §3, §4, §7, §8, §9, §10, §13, §14, §15, §16.1, §16.2, §16.4, §17, §18, §19, §20	Refinement pass: §1 product description rewritten as Reputation Market with t* < T and bandwagon dn/dt absorbed as ideological framing; §3 G1 neutralised, G8 added (propagation-derivable from dataset); §4 personas reframed to Ruling Party / Opposition / Audience (positions in a debate, not identities of people); §7 single-side rule + F-BET-10 opposite-side rejection added; F-BET-8 deleted (structurally impossible under F-AUTH-ADMIN); §8 friendly-fire mechanic added (F-COMMENT-6/7/8) with shared write-rate budget; §9 ranking function locked as open-source AGPL-3.0 in `RANKING.md`, deterministic, universal, ADR-RANKING gating; §13 F-AUTH-ADMIN added (route-gated admin auth, no `users` row, structurally separate from participants), F-AUTH-3 rewritten as auto-assigned permanent pseudonym + PFP from pre-generated identity-pool (DGX Spark + Flux), F-AUTH-4 expanded with inline-scrollable ToS pattern + acceptance evidence, F-AUTH-5 covers both session types; §15 rewritten as Admin Control Centre (hub at `/admin` + inline mod affordances), F-ADMIN-6 highlight tool deleted, three-tab structure (Home / Markets / Moderation); §10.1 / §10.8 / §3 G7 phrasing aligned with structural admin-as-not-a-user	End-to-end refinement session covering thesis framing, persona model, friendly-fire mechanic, ranking function, single-side rule, identity assignment, ToS workflow, admin architecture restructure.	—

DATE	VERSION	SECTION	CHANGE	RATIONALE	ADR
			separation; §16.1 OTP rate-limit constants added; §16.4 audit log catalogue extended with `friendly_fire_events`, `sessions`, `admin_sessions`, `identity_pool`; §17 acceptance tests rebuilt (~50 new test rows, ~5 stale rows removed); §18 expanded with new out-of-scope items covering personalised ranking, engagement-metric ranking inputs, online-learned weights, session expiry, fresh-OTP gates, multi-device session lists, account-recovery, in-product analytics export, public mod log, admin role flag; §19 Q13–Q18 added covering ADR-RANKING, friendly-fire fine-tuning, ADR-PSEUDONYM, OTP rate-limit values, ADR-TOS-UPDATE, ADR-ADMIN-AUTH.		

Appendix A AI moderation category-to-track mapping

Source vendors: OpenAI moderation API (text), PhotoDNA-or-equivalent (image hash for CSAM), general image classifier TBD (adult / violence / weapons).

AI CATEGORY	SOURCE LAYER	TRACK	NOTES
`csam` (hash match)	Image — PhotoDNA	A	Auto-block + auto-report + auto-ban. Legal floor.
`sexual/minors`	Text — OpenAI	A	Auto-block + auto-report + auto-ban.
`sexual`	Text — OpenAI	A	Auto-block + auto-ban. No product fit.
`nsfw` / adult imagery	Image — classifier	A	Auto-block + auto-ban. No product fit.
`violence/graphic`	Text — OpenAI	B	Admin review. Edges: war markets, journalistic context.
`violence` (image)	Image — classifier	B	Admin review.
`harassment`	Text — OpenAI	B	Admin review.
`harassment/threatening`	Text — OpenAI	B	Admin review. Threats specifically.
`hate`	Text — OpenAI	B	Admin review. Edges: quoting slurs to criticise.
`hate/threatening`	Text — OpenAI	B	Admin review.
`self-harm`	Text — OpenAI	B	Admin review.
`weapons`	Image — classifier	B	Admin review. Edges: weapon-policy markets.
(below threshold)	—	C	Posts normally.

Specific confidence thresholds per category per track → number-tuning pass after Aug 15–31 sample-content testing.

Appendix B Constants scaffold for the number-tuning pass

This appendix is the structural placeholder for the number-tuning pass. Each constant has a `value` field set to `TBD` until that pass completes. The number-tuning pass produces an ADR documenting the values and the test data they were tuned against.

```
INITIAL_USER_DHARMA = TBD
ADMIN_INITIAL_DHARMA = TBD
DAILY_ALLOWANCE_DHARMA = TBD
POOL_SEED_PER_MARKET_DEFAULT = TBD
BET_MIN_STAKE = TBD
COMMENT_MAX_LENGTH = TBD
REPLY_DEPTH_MAX = TBD
SLIPPAGE_WARNING_PCT_THRESHOLD = TBD
IN_FLIGHT_BET_TIMEOUT_SEC = TBD
POLL_INTERVAL_MS_DEBATE_VIEW = TBD
OTP_TTL_MIN = TBD
OTP_REQUESTS_PER_EMAIL_PER_HOUR = TBD
OTP_REQUESTS_PER_IP_BURST_PER_MIN = TBD
RATE_LIMIT_PER_MARKET_PER_DAY = TBD
RATE_LIMIT_BURST_PER_MIN = TBD
AI_FLAG_THRESHOLD_TRACK_A_CSAM = TBD            # likely effectively zero — any hash
match
AI_FLAG_THRESHOLD_TRACK_A_NSFW = TBD
AI_FLAG_THRESHOLD_TRACK_A_SEXUAL_MINORS = TBD
AI_FLAG_THRESHOLD_TRACK_B_VIOLENCE = TBD
AI_FLAG_THRESHOLD_TRACK_B_THREATS = TBD
AI_FLAG_THRESHOLD_TRACK_B_HATE = TBD
AI_FLAG_THRESHOLD_TRACK_B_HARASSMENT = TBD
```

End SPEC.1 v1.0-draft.